Classical Field Theory — Exercise Sheet 1

Local actions

Questions only

Use Lecture 1's sign $E_a = \partial L/\partial\phi^a - \partial_\mu p_a^\mu$. All fields and prescribed sources in this sheet are smooth. The first three problems use only differentiation, integration by parts and the fundamental lemma. The final problem also uses finite-dimensional linear ODEs.

Exercise 1.1 (A complex source and two real variations). On $\mathbb{R}^{1,1}$ let $\Phi = \phi_1 + \mathrm{i}\phi_2$ be a complex field, $J = J_1 + \mathrm{i}J_2$ a prescribed complex source, and $m \geq 0$. The density is

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\Phi_x|^2 - m^2|\Phi|^2\big) + \mathrm{Re}(\bar{J}\Phi).$$

For the action, take any compact region $D \subset \mathbb{R}^{1,1}$ and variations H whose support lies in the interior of D.

(a) Compute both real momentum currents and Euler expressions. Express the stationarity condition as one complex equation and derive the first variation for a compactly supported complex variation H.

(b) Find the source for which $\Phi(t,x) = (t^2+1) + \mathrm{i}(t^3 - x^2)$ is stationary.

(c) If only real-valued variations H are tested, must the complex equation hold? Give a counterexample with $m = 0$ and $J = 0$.

Exercise 1.2 (Natural conditions at the two spatial ends). Let $T, \ell > 0$, $m \geq 0$ and $\beta_0, \beta_\ell \geq 0$. On the rectangle $D = [0,T] \times [0,\ell]$, a real field has action

$$S_D[\phi] = \int_0^T\!\!\int_0^\ell \left(\frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - \frac{1}{2}m^2\phi^2 + J\phi\right)\mathrm{d}x\mathrm{d}t - \frac{1}{2}\int_0^T \big(\beta_0\phi(t,0)^2 + \beta_\ell\phi(t,\ell)^2\big)\mathrm{d}t,$$

where J is prescribed. Variations vanish near $t = 0, T$ and are unrestricted at $x = 0, \ell$.

(a) Integrate coordinate by coordinate and prove that stationarity is equivalent to the bulk equation and two Robin boundary conditions. Include the case $\beta_0 = \beta_\ell = 0$.

(b) For $\ell = 1$, $m = 0$, $J = 2$ and $\beta_0 = \beta_\ell = 1$, find the unique time-independent stationary field.

Exercise 1.3 (Lorentz changes of coordinates). The Lorentz group in two coordinates is the matrix group

$$O(1,1) = \{\Lambda \in GL(2,\mathbb{R}) : \Lambda^{\mathsf{T}}\eta\Lambda = \eta\}, \qquad \eta = \mathrm{diag}(1,-1).$$

For $|v| < 1$, set

$$\gamma_v = (1-v^2)^{-1/2}, \qquad \begin{pmatrix} t' \\ x' \end{pmatrix} = \Lambda_v \begin{pmatrix} t \\ x \end{pmatrix}, \qquad \Lambda_v = \gamma_v \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}.$$

(a) Verify that $\Lambda_v \in O(1,1)$ and $\det \Lambda_v = 1$.

(b) For a scalar field, set $\phi'(t',x') = \phi(t,x)$ at corresponding points. Use the chain rule to prove invariance of $\partial_t^2 - \partial_x^2$ and of $\frac{1}{2}(\phi_t^2 - \phi_x^2) - V(\phi)$, where V is any smooth function. Conclude equality of the scalar actions on corresponding compact regions.

Exercise 1.4 ⋆. Let $N \geq 3$, $h_{\mathrm{lat}} = 2\pi/N$, and let $q_j(t)$, $j = 0, \dots, N-1$, be real coordinates with periodic indexing. The chain Lagrangian is

$$L_h = h_{\mathrm{lat}} \sum_{j=0}^{N-1} \left[\frac{1}{2}\dot{q}_j^2 - \frac{1}{2h_{\mathrm{lat}}^2}(q_{j+1} - q_j)^2 - \frac{1}{2}m^2 q_j^2 \right], \qquad m \geq 0.$$

(a) Derive its equations by varying every q_j.

(b) Let $T > 0$ and let ϕ be a given smooth periodic solution of $\phi_{tt} - \phi_{xx} + m^2\phi = 0$ on $[0, T] \times \mathbb{T}$. Assume $M_4 \geq 0$ and $|\partial_x^4 \phi| \leq M_4$ there. Prove that sampling this field in the chain equation produces a residual bounded by $h_{\mathrm{lat}}^2 M_4/12$ at every node.

(c) Give the chain the exact sampled initial position and velocity. Set $e_j = q_j - \phi(t, jh_{\mathrm{lat}})$, $D_h e_j = (e_{j+1} - e_j)/h_{\mathrm{lat}}$ and $\|w\|_h^2 = h_{\mathrm{lat}} \sum_j w_j^2$. With $A_h = \sqrt{2\pi} M_4 h_{\mathrm{lat}}^2/12$, prove, for $0 \leq t \leq T$,

$$\left(\|\dot{e}(t)\|_h^2 + \|D_h e(t)\|_h^2 + m^2 \|e(t)\|_h^2 \right)^{1/2} \leq A_h t, \qquad \|e(t)\|_h \leq \frac{1}{2} A_h t^2.$$

The finite linear chain has a global solution by the linear ODE theorem. No existence theorem for the continuum equation is assumed beyond the given ϕ.

Classical Field Theory — Exercise Sheet 2

Noether currents and their limits

Questions only

Use Lecture 2's signs $j^\mu = p_a^\mu Q^a - B^\mu$ and $\partial_\mu j^\mu = -E_a Q^a$. All fields and coefficients below are smooth. The unindexed $Q(t)$ denotes an integrated charge, and Q^a denotes a characteristic. The last problem is a different field theory from Klein–Gordon.

Exercise 2.1 (An improvement can change a boundary charge). Let j^μ be a current in $1+d$ dimensions and let $K^{\mu\nu} = -K^{\nu\mu}$ be smooth. Define $\widetilde{j}^\mu = j^\mu + \partial_\nu K^{\mu\nu}$. An added current with identically zero divergence is called an *identically conserved improvement* in this problem.

(a) Prove that the two currents have the same divergence for every field. Show that their charges agree on the spatial torus when K is periodic. On $\mathbb{R}^d$, assume the original slice charge exists and show equality when $K^{0i}(t,\cdot)$ is compactly supported.

(b) In $1+1$ dimensions take $K^{01} = t/(1+x^2)$ and $K^{10} = -K^{01}$. Compute the added density and flux. Find the change in total charge on $\mathbb{R}$ and the change in charge on the fixed interval $[0,1]$. Does an identically conserved improvement leave every regional charge unchanged?

Exercise 2.2 (The converse with an off-shell identity supplied). For a first-order density, suppose a legitimate characteristic Q^a and specified local current j^μ satisfy $\partial_\mu j^\mu = -E_a Q^a$ for arbitrary fields, before imposing their equations. Prove that $B^\mu = p_a^\mu Q^a - j^\mu$ gives $\delta_Q L = \partial_\mu B^\mu$.

Apply this calculation on $\mathbb{R}^{1,1}$ to $L = \frac{1}{2}(\phi_t^2 - \phi_x^2)$, $\eta(t,x) = t + x$, and

$$j^0 = \eta\phi_t - \phi, \qquad j^1 = -\eta\phi_x + \phi.$$

Find its characteristic, the divergence term and an actual one-parameter point transformation realising that characteristic. Verify the off-shell identity directly.

For context, the following broader correspondence is a *cited input*, not a task to prove: in the local finite-derivative formulation for the scalar wave Euler–Lagrange equation, conservation-law classes have generalized variational characteristics. Currents are identified modulo currents vanishing on solutions and identically conserved improvements; characteristics are identified modulo ones vanishing on solutions. Here local means depending smoothly on the coordinates, field and finitely many of its derivatives. This is the scalar wave case of Olver, §5.3, Theorem 5.58, p. 334. Explain why the assertion that an arbitrary current is conserved on solutions is not the supplied off-shell identity used in the first calculation.

Exercise 2.3 (A Noether charge escapes to infinity). Let p and q be real fields on $\mathbb{R}_t \times \mathbb{R}_x$, with density $L = p(q_t + x^2 q_x)$. Choose $p_0 \in C_c^\infty((1,2))$, nonnegative and with integral one, and extend it by zero to $\mathbb{R}$. Set $q = 0$ and

$$p(t,x) = \begin{cases} (1+tx)^{-2} p_0\big(x/(1+tx)\big), & 1+tx > 0, \\ 0, & 1+tx \le 0. \end{cases}$$

(a) Derive the two field equations. Find the Noether current for the constant shift of q. Prove that the displayed fields are smooth at every finite spacetime point and satisfy both equations.

(b) Compute $Q(t) = \int_{\mathbb{R}} p(t,x)\mathrm{d}x$. Find its derivative for $t > 0$ and the limiting flux as $x \to +\infty$. Show that the initial charge is one but the charge is zero for $t \ge 1$.

(c) Identify exactly which full-space charge hypothesis in Lecture 2 fails. Why is this neither an off-shell example nor a claim about Klein–Gordon propagation?

Classical Field Theory — Exercise Sheet 3

Stress, improvements and angular momentum

Questions only

Use $T^{\mu}{}_{\nu} = p^{\mu}\partial_{\nu}\phi - \delta^{\mu}_{\nu}L$, $\eta = \mathrm{diag}(1,-1,\dots,-1)$ and the inverse-metric Hilbert convention of Lecture 3. All quantities below are smooth. The first problem has d spatial coordinates; the remaining problems have three. Problems 1–3 use Lectures 1–3. Problem 4 is a return problem after Lecture 9; its electromagnetic component conventions are supplied in the question.

Exercise 3.1 (Work by an external coupling). On $I \times \mathbb{T}^d$ consider

$$L = \frac{1}{2}\phi_t^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{\lambda(t)}{4}\phi^4,$$

where I is an open interval and λ is prescribed and smooth.

(a) Derive the field equation and energy density. For a classical solution, find the derivative of its total energy, including the term due to explicit time dependence. Decide whether its physical spatial momentum $P^i = \int_{\mathbb{T}^d} T^{0i}\mathrm{d}^d x$ remains constant.

(b) Set $I = (0,\pi/2)$, $\lambda(t) = 1/\sin^2 t$ and $\phi(t,x) = \sin t$. Verify the field equation, compute its energy and check the sign and value of the energy derivative by both methods.

Exercise 3.2 (Angular momentum and weighted decay). For the autonomous scalar density $L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$ on $\mathbb{R}^{1+3}$, raise the second stress index with η. For fixed distinct spatial indices i, j, define

$$M^{\mu ij} = x^i T^{\mu j} - x^j T^{\mu i}.$$

(a) Prove its local conservation on solutions, keeping the derivatives of the explicit coordinate factors.

(b) Prove that $J^{ij}(t) = \int_{\mathbb{R}^3} M^{0ij}\mathrm{d}^3x$ is constant if, on every compact time interval, every component of $T^{\mu\nu}$ is bounded by $C(1+|x|)^{-4-\sigma}$ for some $C, \sigma > 0$. Explain why an unweighted integrability statement for stress alone does not automatically establish the angular charge hypotheses.

Exercise 3.3 (Belinfante improvement from a spin identity). Suppose a smooth tensor $T^{\mu\nu}$ on $\mathbb{R}^{1+3}$ is divergence-free, and a supplied smooth tensor $S^{\lambda\mu\nu}$ satisfies

$$S^{\lambda\mu\nu} = -S^{\lambda\nu\mu}, \qquad \partial_\lambda S^{\lambda\mu\nu} = T^{\nu\mu} - T^{\mu\nu}.$$

This spin identity is a given hypothesis; it is not inferred from scalar Lecture 3. Define

$$B^{\lambda\mu\nu} = \frac{1}{2}\big(S^{\lambda\mu\nu} + S^{\mu\nu\lambda} + S^{\nu\mu\lambda}\big), \qquad \Theta^{\mu\nu} = T^{\mu\nu} + \partial_\lambda B^{\lambda\mu\nu}.$$

Prove that Θ is symmetric and divergence-free. Whenever the slice integrals exist, find the difference between its integrated four-momentum and that of T. Give a sufficient boundary condition for equality, and explain why local improvement alone is insufficient.

Exercise 3.4 (Maxwell Hilbert stress, after Lecture 9). Return to this problem after Lecture 9. Let $F_{\mu\nu} = -F_{\nu\mu}$ be held fixed with both indices down while a Lorentz metric is varied. Use the density and Hilbert convention

$$L = -\frac{1}{4} g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma}, \qquad \delta S = \frac{1}{2} \int T_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g, \quad k^{\mu\nu} = \delta g^{\mu\nu}.$$

The volume variation is $\delta \mathrm{vol}_g = -\frac{1}{2} g_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. Derive $T_{\mu\nu}$ by varying both inverse metrics. At $g = \eta$, set $F_{i0} = E_i$ and $F_{ij} = \epsilon_{ijk} B_k$, with $\epsilon_{123} = 1$. Compute T_{00} and T^{i0} in terms of the vectors $\mathbf{E}$ and $\mathbf{B}$. Evaluate them for $\mathbf{E} = (1, 2, 0)$ and $\mathbf{B} = (0, 0, 3)$.

Classical Field Theory — Exercise Sheet 4

Formula, stability and weak derivatives

Questions only

Use the unitary Fourier transform $\widehat{f}(k) = (2\pi)^{-1/2}\int_{\mathbb{R}} e^{-ikx}f(x)\mathrm{d}x$ and $\|f\|_{H^1}^2 = \int_{\mathbb{R}}(1+k^2)|\widehat{f}(k)|^2\mathrm{d}k$. All wave equations on this sheet have one spatial dimension and zero mass. The solution class is that of Theorem 4.3.

Exercise 4.1 (Two initial data and two travelling packets). Let $f, g \in C_c^\infty(\mathbb{R};\mathbb{R})$ be initial displacement and velocity.

(a) Derive d'Alembert's formula from Lecture 4's Fourier multipliers. Use an oriented integral when $t < 0$. Verify the equation and both data, and justify identifying the formula with the energy-class solution.

(b) Let F, G be smooth real functions supported in $[-1/4, 1/4]$ and strictly positive on its interior. Take

$$f(x) = F(x+3) + 2G(x-2), \qquad g(x) = -F'(x+3) + 2G'(x-2).$$

Find the solution. For $t \geq 0$, find the open time intervals on which an observer at $x = 0$ sees each contribution. Where and when do the packet centres meet? Find the corresponding visibility interval at that position.

Exercise 4.2 (Hadamard instability for an elliptic Cauchy problem). For each positive integer n, seek a smooth function $u_n(x,y)$, periodic in $x \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, satisfying

$$\partial_y^2 u_n + \partial_x^2 u_n = 0, \qquad u_n(x,0) = 0, \qquad \partial_y u_n(x,0) = e^{-\sqrt{n}}\cos(nx).$$

Find such a solution by separation of variables. Prove that the initial velocity tends to zero in every fixed $C^r(\mathbb{T})$ seminorm, while $\|u_n(\cdot,y)\|_{L^2(\mathbb{T})} \to \infty$ for every fixed $y > 0$. What continuity claim does this disprove? Why does it not contradict Lecture 4?

Exercise 4.3 (A corner and its weak derivatives ⋆). Let $f(x) = e^{-|x|}$. Compute its unitary Fourier transform and its H^1 norm. Identify its first weak derivative by testing against every $\zeta \in C_c^\infty(\mathbb{R})$, and calculate its second distributional derivative. Decide whether f belongs to the space

$$H^2(\mathbb{R}) = \left\{ h \in L^2(\mathbb{R}) : \int_{\mathbb{R}} (1+k^2)^2 |\widehat{h}(k)|^2 \mathrm{d}k < \infty \right\}.$$

Distinguish the value assigned to a weak derivative at the single point zero from the existence of a classical derivative there.

Exercise 4.4 (The sharp massless sine bound). Choose a real even $\rho \in C_c^\infty((-1,1))$ with $\|\rho\|_2 = 1$. For $0 < \varepsilon \leq 1$, let g_ε satisfy $\widehat{g}_\varepsilon(k) = \varepsilon^{-1/2}\rho(k/\varepsilon)$, and let v_ε have initial data $(0, g_\varepsilon)$.

(a) Prove that g_ε is real and normalized in L^2, and that for every $T > 0$,

$$\sup_{|t|\leq T} \|v_\varepsilon(t) - t g_\varepsilon\|_2 \leq \frac{T^3\varepsilon^2}{6}.$$

(b) For fixed real t, determine the operator norm on real $L^2(\mathbb{R})$ of the Fourier multiplier $S_t(|k|)$, where $S_t(0) = t$. Explain why conservation of massless energy cannot give a bound for $\|v(t)\|_2$ uniform in both time and all initial data with energy $1/2$.

Classical Field Theory — Exercise Sheet 5

Cones with forcing and interior tails

Questions only

Write $\Box = \partial_t^2 - \Delta$. Every cone argument below concerns real C^2 fields on an open neighbourhood of the closed cone $\{0 \le t \le R,\ |x-a| \le R-t\}$, with $R > 0$. No existence theorem for a nonlinear equation is assumed or proved here.

Exercise 5.1 (Source work on a shrinking cone). Suppose $(\Box + m^2)w = F$, with $m \ge 0$ and continuous F. Let $E_m(t)$ be the integral over $B_{R-t}(a)$ of $e_m = (w_t^2 + |\nabla w|^2 + m^2 w^2)/2$.

(a) Derive the exact shrinking-ball balance, including source work and all lateral flux terms. Prove, for $0 \le t < R$,

$$E_m(t) \le e^t \left[E_m(0) + \frac{1}{2} \int_0^t e^{-s} \|F(s)\|^2_{L^2(B_{R-s}(a))} \mathrm{d}s \right].$$

(b) If $F = 0$ on the cone and both initial data vanish on its base, prove that $w = 0$ on the closed cone, including when $m = 0$. Explain why this is a local statement even if total energy is infinite.

Exercise 5.2 (Finite speed for a semilinear equation, conditional on solutions). Let $V \in C^2(\mathbb{R})$ and suppose two real C^2 fields ϕ_1, ϕ_2 solve $\Box\phi + V'(\phi) = 0$ on a neighbourhood of the same closed cone. Assume their displacements and velocities agree on $B_R(a)$ at time zero.

(a) Prove that the fields agree on the closed cone. Use the mass-one energy of their difference, allowing V'' to have either sign. Give an explicit finite coefficient in the energy inequality in terms of a bound for $|V''|$ on the field values occurring in the cone.

(b) Suppose instead that the fields are defined on an open slab $I \times \mathbb{R}^d$, with $0 \in I$, and their two initial data agree outside a closed set K. Prove agreement at every (t, x) with $\mathrm{dist}(x, K) > |t|$. What extra condition makes the zero field an available comparison solution?

Exercise 5.3 (Huygens' principle and the dimension). Consider the massless wave with zero displacement and real $g \in C_c^\infty(\mathbb{R}^d)$ as initial velocity. The following classical representation formulas may be used as given facts for $t > 0$:

$$u_1(t,x) = \frac{1}{2}\int_{x-t}^{x+t} g(y)\,\mathrm{d}y, \qquad u_2(t,x) = \frac{1}{2\pi}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,\mathrm{d}^2y,$$

$$u_3(t,x) = \frac{1}{4\pi t}\int_{|y-x|=t} g(y)\,\mathrm{d}S_y.$$

For the higher-dimensional classification one may use these general forms. Let α_d be the volume of the unit ball, $\fint$ the integral divided by the region's volume or area, and $D_t = t^{-1}\partial_t$. For odd $d \geq 3$ and even $d \geq 2$, respectively,

$$u_d = \frac{1}{1\cdot 3\cdots(d-2)} D_t^{(d-3)/2}\left[t^{d-2}\fint_{|y-x|=t} g(y)\,\mathrm{d}S_y\right],$$

$$u_d = \frac{1}{2\cdot 4\cdots d} D_t^{(d-2)/2}\left[\frac{1}{\alpha_d}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,\mathrm{d}^dy\right].$$

These facts hold for the smooth compact data specified here; see Evans, *Partial Differential Equations*, 2nd ed., §2.4.1, formulas (31), (38), Theorems 2–3, pp. 73–77. Nonzero displacement contributes the time derivative of the corresponding velocity formula with g replaced by that displacement.

(a) For $d = 1, 2, 3$, let $g \geq 0$, $g \not\equiv 0$, with support inside $B_1(0)$. Decide whether $u_d(t,0)$ vanishes for $t > 1$. Explain the difference between finite speed and dependence only on data arbitrarily near the base sphere.

(b) Use the general formulas to classify all positive spatial dimensions. For even d, test data supported strictly inside the base sphere so that every time derivative acts on a smooth kernel. Explain why a statement that all even-dimensional kernels are positive would be too strong.

Classical Field Theory — Exercise Sheet 6

Causal kernels and frequency boundary values

Questions only

Use $\Box = \partial_t^2 - \Delta$ and Lecture 6's normalized kernel $G_{\mathrm{ret}} = \delta(t - |x|)/(4\pi|x|)$ in three spatial dimensions. Its meaning at the vertex is its test pairing, not a product of distributions. All distribution pairings are bilinear. Write H for the Heaviside function; its value at zero does not affect the distributions below.

Exercise 6.1 (An entire radial source response). Let $q \in C_c^\infty(\mathbb{R})$, and let h be a smooth even compactly supported function such that $x \mapsto h(|x|)$ is smooth on $\mathbb{R}^3$. Take $f(t,x) = q(t)h(|x|)$ and $u = G_{\mathrm{ret}} * f$.

(a) Perform the angular integration to obtain a one-dimensional radial integral for $u(t,r)$, $r = |x| > 0$, with an inner integral over travel distance. Find its limit at $r = 0$ and explain why no singularity remains there.

(b) Set $v(t,r) = ru(t,r)$ for $r \geq 0$ and extend it oddly to the real line. Reduce the equation for v to a forced one-dimensional wave and derive its causal double-integral formula. Check its agreement with part (a).

(c) Let $a < b$ and $R > 0$. If $\operatorname{supp} q \subset [a,b]$ and $\operatorname{supp} h \subset [-R,R]$, find the possible response interval at an observer $r > R$. If $q > 0$ on (a,b) and $h > 0$ for $|s| < R$, prove that the response is positive throughout the interior of that interval.

Exercise 6.2 (Which side of the frequency poles?). The spacetime Fourier convention is

$$\widetilde{F}(\nu,k) = (2\pi)^{-2} \int e^{i\nu t - ik\cdot x} F(t,x)\mathrm{d}t\mathrm{d}^3x.$$

For a tempered distribution it is defined by transposition on Schwartz tests, with this same kernel. Fourier transformation preserves Schwartz space and its topology, so this definition preserves convergence of tempered distributions. The Schwartz Fourier automorphism is the one from *Analysis* I used in Lecture 4; transposition here specifies its distributional extension. Let $G_\varepsilon = e^{-\varepsilon t} G_{\mathrm{ret}}$, $\varepsilon > 0$; the product is understood on the future cone, where the factor is bounded.

(a) Compute its spatial transform, then its full spacetime transform. Prove convergence as $\varepsilon \downarrow 0$ on Schwartz tests and express $\widetilde{G}_{\mathrm{ret}}$ as a frequency boundary value. Check the Fourier coefficient of the equation $\Box G_{\mathrm{ret}} = \delta^4$.

(b) For fixed $\kappa = |k| > 0$, derive the principal-value and delta terms of that boundary value. Here $\langle \mathrm{pv}(1/s), \zeta \rangle = \lim_{a \downarrow 0} \int_{|s|>a} \zeta(s)/s \, \mathrm{d}s$. Derive the identity for $(s \pm i0)^{-1}$ by separating real and imaginary parts. Compare the advanced and half-retarded plus half-advanced kernels. State why the fixed-κ decomposition cannot simply be evaluated at $\kappa = 0$.

Exercise 6.3 (Descending the spatial dimension). Starting with the three-dimensional retarded pairing, integrate out one auxiliary spatial coordinate to construct a kernel G_2 on $\mathbb{R}^{1+2}$, and two auxiliary coordinates to construct G_1 on $\mathbb{R}^{1+1}$. Make the construction legitimate on compact smooth tests: a test independent of the auxiliary coordinates is not itself compactly supported. Derive explicit locally integrable formulas, prove $(\partial_t^2 - \Delta_d)G_d = \delta^{1+d}$ for $d = 1, 2$, and identify their support and their relation to Sheet 5's interior tails.

Classical Field Theory — Exercise Sheet 7

Modes, complex pairs and admissibility

Questions only

Use $\{F,G\} = \int (F_\phi G_\pi - F_\pi G_\phi)$ and $\Omega = \int \delta\phi \wedge \delta\pi$. Except for the explicitly finite torus model in Problem 1, functionals have Lecture 7's smooth Schwartz-valued gradients. A formal distributional derivative does not automatically belong to that class.

Exercise 7.1 (Real oscillators in a finite torus model). On $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, fix an integer $N \geq 2$ and use the real basis

$$e_0 = (2\pi)^{-1/2}, \qquad e_{n,c} = \pi^{-1/2}\cos(nx), \qquad e_{n,s} = \pi^{-1/2}\sin(nx), \quad 1 \leq n \leq N.$$

Let $\phi = q_0 e_0 + \sum_{n=1}^{N}(q_{n,c}e_{n,c} + q_{n,s}e_{n,s})$ and let π have the same expansion with coefficients p.

(a) Verify orthonormality and compute the pulled-back canonical form and the Hamiltonian $H = \frac{1}{2}\int_{\mathbb{T}}(\pi^2 + \phi_x^2 + m^2\phi^2)\mathrm{d}x$, where $m \geq 0$. Derive the oscillator equations and treat the zero mode at $m = 0$.

(b) Define $\widehat{\phi}_n = (2\pi)^{-1/2}\int e^{-inx}\phi\mathrm{d}x$, and similarly $\widehat{\pi}_n$, for $|n| \leq N$. Express them in the real coordinates and compute their complex bilinear brackets, including $n = 0$.

(c) Compute all nonzero real coefficients and the energy of the data $\phi = 3\cos(2x) - 2\sin x$, $\pi = 5\sin(2x) + \cos x$.

Exercise 7.2 (A complex scalar is two real canonical pairs). Write $\Phi = \phi_1 + i\phi_2$, with both real fields Schwartz, and take

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\nabla\Phi|^2 - m^2|\Phi|^2\big), \qquad m \geq 0.$$

Derive the two momenta and Hamiltonian by real variation. Compute all functional derivatives of $Q = \int(\phi_1\pi_2 - \phi_2\pi_1)\mathrm{d}^d x$ and its action on both fields and both momenta. Identify its finite flow, prove $\{Q,H\} = 0$, and verify admissibility in the product of the two real Schwartz phase spaces. Evaluate Q on $(\phi_1,\phi_2;\pi_1,\pi_2) = (f, 2f; -f, 3f)$, where $f \in \mathcal{S}(\mathbb{R}^d;\mathbb{R})$ and $\|f\|_2 = 1$.

Exercise 7.3 (A smooth window and a sharp interval). In one dimension let $w \in C_c^\infty(\mathbb{R};\mathbb{R})$ be a fixed window and set

$$F_w[\phi,\pi] = \frac{1}{2}\int_{\mathbb{R}} w(x)\phi_x(x)^2\mathrm{d}x, \qquad F_\#[\phi,\pi] = \frac{1}{2}\int_0^1 \phi_x(x)^2\mathrm{d}x.$$

Compute both derivatives against arbitrary real Schwartz variations. Prove F_w is admissible. Is $F_\#$ an admissible functional on the whole Schwartz phase space? Test the field $\phi(x) = (x+x^2)\eta(x)$, where $\eta \in C_c^\infty(\mathbb{R})$ equals one on a neighbourhood of $[0,1]$. Display its endpoint distribution terms.

Exercise 7.4 (The sign of the Hamiltonian commutator). For admissible F, G, let $H_F = D(\operatorname{grad} F)$ and $\mathbb{J}(a,b) = (b,-a)$ as in Lecture 7. Define the vector-field bracket by

$$[X,Y](z) = DY(z)X(z) - DX(z)Y(z),$$

the sign for which it acts on functionals as $X(YK) - Y(XK)$. Use the actual bracket-gradient formula of Lecture 7 to prove $X_{\{F,G\}} = -[X_F, X_G]$. For a nonzero example, choose real $f \in \mathcal{S}$ with $\|f\|_2 = 1$, put $q = \int f\phi$, $p = \int f\pi$, and take $F = q^2/2$, $G = p^2/2$. Compute both vector fields, their commutator, and $X_{\{F,G\}}$.

Classical Field Theory — Exercise Sheet 8

Slices, boundary conditions and causal brackets

Questions only

For real Klein–Gordon variations with compact smooth Cauchy data, use $\omega^\mu(h_1, h_2) = h_1\partial^\mu h_2 - h_2\partial^\mu h_1$. The future-oriented graph element is $(1, -\nabla\tau)\mathrm{d}^d x$. All fields below are smooth. The last problem explicitly restricts the field equation to the massless wave in $3+1$ dimensions.

Exercise 8.1 (An unbounded Cauchy graph). Let $\tau(x) = t_* + v \cdot x$ on $\mathbb{R}^d$, where $t_* \in \mathbb{R}$ and $v \in \mathbb{R}^d$, $|v| < 1$. Let $h_1, h_2 \in \mathcal{S}_m$ be the actual compact-Cauchy-data solutions of Lecture 8, $m \geq 0$.

(a) Prove that the graph is spacelike and meets each inextendible timelike curve exactly once. A future timelike curve can be parametrized by coordinate time with $|\dot{x}(t)| < 1$; justify that its time range is all of $\mathbb{R}$. Here an inextendible curve has no past or future endpoint in Minkowski spacetime.

(b) Although τ is unbounded, prove that the current is supported in a compact subset of the region between the graph and $t = 0$. Give an explicit spatial bound in terms of t_*, $|v|$, and a radius R containing both sets of initial data. Use signed time integration to prove $\Omega_{\Sigma_\tau}(h_1, h_2) = \Omega_0(h_1, h_2)$, including where the graph crosses zero.

(c) Choose nonzero real $f \in C_c^\infty(\mathbb{R}^d)$ and solutions with data $(f, 0)$ and $(0, f)$. Determine their pairing on this graph.

For comparison, the following general result is supplied: in full Minkowski spacetime, the same compact-data solutions have finite, equal symplectic integrals on every smooth spacelike Cauchy surface with future orientation. It uses spacelike compact support, not just current closure. See Khavkine, §3.3.3, Lemma 3.11, p. 41, arXiv:1402.1282. Your proof here should establish the affine case directly.

Exercise 8.2 (Common boundary conditions remove side flux). Let two real solutions of $(\partial_t^2 - \partial_x^2 + m^2)h = 0$ be smooth on a neighbourhood of $I \times [0, L]$, where I is an open interval, $L > 0$ and $m \geq 0$. Set $\Omega_{[0,L]}(h_1, h_2) = \int_0^L (h_1 h_{2,t} - h_2 h_{1,t})\mathrm{d}x$. At each endpoint, consider the following common conditions on both solutions: Dirichlet $h = 0$, Neumann $\partial_n h = 0$, or real Robin $\partial_n h + \kappa h = 0$, with the same prescribed real κ for both solutions at that endpoint. The two endpoints may have different such conditions.

(a) Prove that each choice makes the side symplectic flux zero and conserves $\Omega_{[0,L]}$. Specify the outward derivative at each endpoint.

(b) For $m = 0$, $L = \pi$, compare $h_D(t, x) = \sin x \cos t$ and $h_N(t, x) = 1$. Each obeys one of the listed conditions, but they obey different conditions. Compute their current, pairing and net outward side flux. Is individual satisfaction of some listed condition sufficient?

Exercise 8.3 (A causal source bracket and its sign). In $3+1$ dimensions let G_{ret} be Lecture 6's massless kernel, let $G_{\mathrm{adv}}(t,x) = G_{\mathrm{ret}}(-t,x)$, and put $E = G_{\mathrm{adv}} - G_{\mathrm{ret}}$. For real $f, g \in C_c^\infty(\mathbb{R}^{1+3})$, define on the massless compact-data solution space

$$F_f(\phi) = \int_{\mathbb{R}^{1+3}} f\phi \, \mathrm{d}t\mathrm{d}^3x, \qquad Ef = (G_{\mathrm{adv}} - G_{\mathrm{ret}}) * f.$$

The identification of the resulting causal bracket with the linear *Peierls bracket* may be taken as given; see Khavkine, §§3.3.4–3.3.5, Theorem 3.2, and the scalar case §4.1, arXiv:1402.1282. The signs in this problem must be fixed by the following calculation, using the course convention $\iota_{X_F}\Omega = DF$.

(a) Show that Ef is a smooth homogeneous solution with compact smooth data on every constant slice. Apply the sourced current balance to $G_{\mathrm{ret}} * f$ and a homogeneous compact-data solution h to prove $\Omega(Ef, h) = \int fh$. Identify X_{F_f} and derive the bracket $\{F_f, F_g\} = \int f(Eg)$. Prove its antisymmetry and its vanishing when the two test supports are spacelike separated.

(b) As a separate finite-dimensional sign test, take one oscillator $\ddot{q} + \nu^2 q = 0$, $\nu \geq 0$, with $\{q(0), p(0)\} = 1$ and $p = \dot{q}$. Compute $\{q(t), q(s)\}$ and $\{q(t), p(s)\}$, using continuous limits at $\nu = 0$. Recover the equal-time canonical sign. Explain why this finite model does not make point evaluation of a field admissible in Lecture 7.

Classical Field Theory — Exercise Sheet 9

Gauge identities and source conditions

Questions only

Use $A_{\mathrm{em}} = -\phi_{\mathrm{em}}\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}x$, $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t\mathbf{A}$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$. Unless a weaker class is explicitly discussed, fields and coefficients are smooth. Compact support is stated separately in each problem.

Exercise 9.1 (The adjoint differentiates coefficients too). Let a smooth first-order density $L(x,\phi,\partial\phi)$ have Euler expressions E_a on an open coordinate region $U \subset \mathbb{R}^{1+d}$. Vary its action on a compact region $D \subset U$, with every parameter supported in the interior of D. For smooth parameters ϵ^α, consider the finite-order variation

$$\delta_\epsilon\phi^a = \sum_{|I|\le r} R^{aI}_\alpha(x, j^s\phi)\partial_I\epsilon^\alpha.$$

Here I is a derivative multi-index, each multi-index occurs once in the sum, and $j^s\phi$ denotes the field and all its derivatives through finite order s. The coefficients are smooth functions of these arguments. Assume the action is invariant under these variations for every compact parameter and every smooth field, without imposing the field equations.

(a) Derive the identity satisfied by the Euler expressions, including the variable coefficients. For the one-dimensional test operator $R\epsilon = (1+x^2)\epsilon_x + x\epsilon$, display R^*E explicitly. This test asks for its adjoint, not for a claim that every density has that symmetry.

(b) For Maxwell's transformation $\delta\phi_{\mathrm{em}} = -\epsilon_t$, $\delta\mathbf{A} = \nabla\epsilon$, derive the identity for the scalar and vector Euler expressions. With $\mathcal{M}_0 = \operatorname{div}\mathbf{E}$ and $\boldsymbol{\mathcal{M}} = \partial_t\mathbf{E} - \operatorname{curl}\mathbf{B}$, include a prescribed source $(\rho,\mathbf{J})$ and identify its necessary conservation condition. Explain the distributional interpretation for only C^2 potentials.

Exercise 9.2 (Polarization and magnetization). Let $P, M \in C^\infty_c(\mathbb{R}^{1+3};\mathbb{R}^3)$ and define

$$\rho = -\operatorname{div}P, \qquad \mathbf{J} = \partial_t P + \operatorname{curl}M.$$

Prove conservation, zero total charge on each slice and $\int y\rho(t,y)\mathrm{d}^3y = \int P(t,y)\mathrm{d}^3y$. For any smooth potential, rewrite the integrated source coupling $\int(-\rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A})\mathrm{vol}$ in terms of P, M and the fields, displaying why all discarded terms vanish. Does this construction assert that the potential of this source is unique?

Exercise 9.3 (Vacuum duality with the Lorentzian star). For a vacuum Maxwell field, put $F' = \star F_{\mathrm{em}}$, using exactly the Hodge table of Lecture 9 and $\star^2 = -1$ on two-forms. Find the transformed electric and magnetic vectors, and verify all four vacuum component equations. Determine the effect of two and four successive transformations. Check the energy and Poynting densities for the constant vacuum fields $\mathbf{E} = (1,0,2)$, $\mathbf{B} = (0,3,0)$ before and after the map. These are densities, not finite total energies on $\mathbb{R}^3$. Does the same transformation preserve Maxwell's equations with a prescribed nonzero electric current three-form $\mathcal{J}$ and no magnetic source?

Exercise 9.4 (Conservation does not remove a time-face term). Let $D = [t_0, t_1] \times \mathbb{T}^3$, $t_1 > t_0$, and use the Maxwell density $L = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) - \rho\phi_{\mathrm{em}} + \mathbf{J} \cdot \mathbf{A}$. Take smooth periodic fields and a prescribed smooth conserved source. Allow a smooth periodic gauge function $\chi(t, x)$ with nonzero time-face values.

(a) Find the exact action change, retaining the time faces and explaining the cancellation on opposite spatial faces.

(b) Set $\rho = \sin x^1$, $\mathbf{J} = 0$, $\phi_{\mathrm{em}} = \sin x^1$, $\mathbf{A} = 0$, and $\chi = \gamma t \sin x^1$, where real $\gamma \neq 0$. Verify Maxwell's equations before and after the transformation and compute the nonzero action change. Why is this consistent with Lecture 9's compact-parameter gauge invariance?

Classical Field Theory — Exercise Sheet 10

Residual gauges and charged tails

Questions only

Write $\Box = \partial_t^2 - \Delta$. Use $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t\mathbf{A}$, $\mathbf{B} = \operatorname{curl}\mathbf{A}$ and $\ell = -(\partial_t\phi_{\mathrm{em}} + \operatorname{div}\mathbf{A})$. Lecture 10's theorem assumes a smooth conserved source of compact *spacetime* support and compatible compact smooth potential data. Problems 2–3 explicitly separate a charged background outside that class.

Exercise 10.1 (All residual shifts of Cauchy data). Let a smooth Lorenz-gauge Maxwell potential have initial data $(a_\phi, b_\phi, a_A, b_A)$ as in Lecture 10, with a fixed source ρ. Let χ be the smooth wave with arbitrary real compact smooth data $\chi(0) = p$, $\chi_t(0) = q$. Set $\phi'_{\mathrm{em}} = \phi_{\mathrm{em}} - \chi_t$, $\mathbf{A}' = \mathbf{A} + \nabla\chi$.

(a) Find the increments of all four potential data. Verify directly that both compatibility expressions $b_\phi + \operatorname{div} a_A$ and $-\Delta a_\phi - \operatorname{div} b_A - \rho(0)$ are unchanged. Compare the two initial electromagnetic fields.

(b) Prove that fixing all four potential data rules out every nonzero residual gauge wave in this compact-data class. Contrast this with fixing only the electromagnetic fields. If compactness of the gauge data is dropped, explain why a constant χ is physically trivial.

Exercise 10.2 (A smooth charge with a Coulomb tail). Choose a radial nonnegative $b \in C_c^\infty(\mathbb{R}^3)$ with $\int b\,\mathrm{d}^3x = 1$ and support in $\overline{B_R(0)}$, $R > 0$. Fix real $Q \neq 0$ and prescribe the static source $\rho_Q(x) = Qb(x)$, $\mathbf{J} = 0$ for all time. For $r > 0$ define, writing $\rho_Q(s)$ for its radial profile,

$$M_Q(r) = 4\pi \int_0^r s^2 \rho_Q(s)\mathrm{d}s, \quad E_r(r) = \frac{M_Q(r)}{4\pi r^2}, \quad \phi_Q(r) = \int_r^\infty E_r(s)\mathrm{d}s.$$

(a) Prove that $E_Q(x) = E_r(|x|)x/|x|$ and $\phi_Q(|x|)$ extend smoothly across zero. Prove Gauss's law there as well as away from zero. Show that $(\phi_{\mathrm{em}}, \mathbf{A}) = (\phi_Q, 0)$ solves Maxwell in Lorenz gauge.

(b) Determine the exact potential and field for $r \geq R$. Prove $\phi_Q \notin L^2$, hence $\phi_Q \notin H^1$, while $\nabla\phi_Q \in L^2$ and the field energy is finite. Calculate the electric energy outside B_R.

(c) Identify the failed hypotheses of Lecture 10's theorem. Why does conservation alone not force $Q = 0$ for this source?

Exercise 10.3 (An affine charged evolution class). Keep the background of Exercise 10.2. Let $(\rho_1, J_1) \in C_c^\infty(\mathbb{R}^{1+3}; \mathbb{R} \times \mathbb{R}^3)$ be conserved. Give compact smooth perturbation data $(a_{1\phi}, b_{1\phi}, a_{1A}, b_{1A})$ satisfying

$$b_{1\phi} + \operatorname{div} a_{1A} = 0, \qquad -\Delta a_{1\phi} - \operatorname{div} b_{1A} = \rho_1(0).$$

Construct a smooth Lorenz-gauge solution for total source $(\rho_Q + \rho_1, J_1)$ and total initial data obtained by adding the static background. Prove its total charge is Q, its electric and magnetic fields have finite energy on each slice, and $\phi_{\mathrm{em}}(t) \in \phi_Q + H^1(\mathbb{R}^3)$ while $\phi_{\mathrm{em}}(t) \notin H^1$. State the uniqueness conclusion actually supplied by Lecture 10. Explain why this construction does not put the charged background in a Schwartz canonical phase space.

Exercise 10.4 (Local potentials with no global real one-form). On the oriented unit sphere, with area form $\sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi$, let $g_{\mathrm{mag}} \neq 0$ be real. On $U_N = S^2 \setminus \{\text{south pole}\}$ and $U_S = S^2 \setminus \{\text{north pole}\}$, respectively, take

$$A_N = g_{\mathrm{mag}}(1 - \cos\theta)\mathrm{d}\varphi, \qquad A_S = -g_{\mathrm{mag}}(1 + \cos\theta)\mathrm{d}\varphi.$$

These are spatial real one-form potentials; the task uses exterior calculus from *Topology and Geometry*, Part I, not bundle theory as a prerequisite.

(a) Verify smoothness at each included pole, compute the common curvature $F = \mathrm{d}A_N = \mathrm{d}A_S$, and find its total flux.

(b) Find a gauge function on a simply connected azimuthal sector in the overlap. Prove that there is no single-valued real gauge function for this shift on the whole overlap, and no globally smooth real one-form A on S^2 with $\mathrm{d}A = F$.

The interpretation by nontrivial bundles and single-valued charged transition phases, including the allowed flux quantization, is a later topic: *Topology and Geometry*, Lectures 20–22. No bundle construction or quantization theorem is assumed in this exercise.

Classical Field Theory — Exercise Sheet 11

Constraints and boundaries

Questions only

Use Lecture 11's canonical bracket, with $\boldsymbol{\Pi} = -\mathbf{E}$. A primary constraint is imposed only after forming the Hamilton equations. All field data below are real and smooth. The second problem uses Schwartz data on $\mathbb{R}^3$; the third explicitly changes the spatial region and its allowed boundary variations.

Exercise 11.1 (An entire constraint chain). Consider two mechanical coordinates $q(t)$ and $u(t)$ with Lagrangian

$$L(q, u, \dot{q}, \dot{u}) = \frac{1}{2}(\dot{q} - u)^2.$$

Use the following consistency procedure: find the image of the Legendre map, form the canonical Hamiltonian on that image, add an arbitrary multiplier for each primary constraint, and preserve the constraints until no new condition appears. The bracket on $(q, u; p_q, p_u)$ is the ordinary canonical bracket of mechanics.

(a) Carry out every step. State which conditions hold strongly and which hold only on the constraint surface. Classify the constraints.

(b) Find a time-dependent generator producing $\delta q = \epsilon(t)$ and $\delta u = \dot{\epsilon}(t)$ for an arbitrary smooth real function ϵ. Determine the transformation of the total-Hamiltonian multiplier needed to map solutions to solutions.

(c) At a fixed time, show that every two points on the constraint surface are gauge equivalent. How many canonical pairs remain on this quotient?

Exercise 11.2 (A mass term changes the constraint class). On $\mathbb{R}^{1+3}$ take the vacuum density

$$L_M = \frac{1}{2}|\mathbf{A}_t + \nabla\phi_{\mathrm{em}}|^2 - \frac{1}{2}|\operatorname{curl}\mathbf{A}|^2 + \frac{M^2}{2}(\phi_{\mathrm{em}}^2 - |\mathbf{A}|^2), \qquad M > 0.$$

At each time all canonical fields and the multiplier are Schwartz. Use the extended bracket of Lecture 11 and define $P[\eta] = \int \eta\pi_\phi \, \mathrm{d}^3x$ for real Schwartz η.

(a) Derive the momenta, canonical Hamiltonian and consistency chain. Does preserving the secondary constraint leave $\partial_t\phi_{\mathrm{em}}$ arbitrary?

(b) Put $\mathcal{D}_M = \operatorname{div}\boldsymbol{\Pi} - M^2\phi_{\mathrm{em}}$ and $C_M[\chi] = -\int \chi\mathcal{D}_M \, \mathrm{d}^3x$. Compute all primary/secondary brackets. For smearings with $\int \eta\chi = 1$, write the two-by-two off-diagonal bracket block and its inverse. Why does its inverse have no massless limit?

(c) Impose both constraints and express the scalar variables in terms of $\mathbf{A}$ and $\boldsymbol{\Pi}$. Pull back the canonical form and determine how many spatial component pairs remain. Contrast this with Maxwell's gauge quotient.

Exercise 11.3 (A boundary generator need not vanish). Let $\Omega = B_1(0) \subset \mathbb{R}^3$. For this problem the phase variables $\mathbf{A}, \mathbf{\Pi}$ are smooth on a neighbourhood of $\overline{\Omega}$, with no fixed normal trace for $\mathbf{\Pi}$. Their canonical form is $\int_\Omega \delta A_i \wedge \delta \Pi_i \, \mathrm{d}^3x$. A functional has a *smooth bulk derivative* if its variation is an integral of smooth coefficients against arbitrary smooth field variations, without a boundary evaluation term. For a fixed smooth real χ, set

$$C_\Omega[\chi] = -\int_\Omega \chi \operatorname{div} \mathbf{\Pi} \, \mathrm{d}^3x, \qquad G_\Omega[\chi] = \int_\Omega \mathbf{\Pi} \cdot \nabla\chi \, \mathrm{d}^3x.$$

(a) Derive their relation, including the surface integral, and vary it. Explain why $C_\Omega[\chi]$ generally lacks a smooth bulk derivative when χ is nonzero on the boundary, while $G_\Omega[\chi]$ has one.

(b) Find the Hamiltonian vector of $G_\Omega[\chi]$. On the Gauss surface $\operatorname{div} \mathbf{\Pi} = 0$, must its value and its contraction with the constrained canonical form vanish?

(c) Evaluate both functionals and their boundary difference for $\mathbf{\Pi} = e_1$ and $\chi = x^1$. Identify the boundary assumption that distinguishes this problem from the all-space integrations by parts in Lecture 11.

Classical Field Theory — Exercise Sheet 12

Completing transverse reduction

Questions only

Use the unitary Fourier transform of Lecture 4. The real canonical form is

$$\Omega((a,p),(b,q)) = \int (a\cdot q - b\cdot p).$$

The projector is $P_{\mathrm{T}}(k) = I - kk^{\mathsf{T}}/|k|^2$ for $k \neq 0$. All claims about a fixed annulus below concern the data and smearings specified in Lecture 12, not pointwise Fourier observables.

Exercise 12.1 (A finite Dirac bracket and an oblique mode). For a fixed $k \in \mathbb{R}^3 \setminus \{0\}$, consider the finite canonical model $(a,p) \in \mathbb{R}^3 \times \mathbb{R}^3$ with $\{a_i, p_j\} = \delta_{ij}$. Set $c_1 = k\cdot p$ and $c_2 = k\cdot a$. Given an invertible constraint matrix $C_{ab} = \{c_a, c_b\}$, its *Dirac bracket* is

$$\{F,G\}_D = \{F,G\} - \{F,c_a\}(C^{-1})_{ab}\{c_b,G\}.$$

This formula is supplied for this finite system.

(a) Compute C, its inverse, and every coordinate Dirac bracket. Check that each constraint has zero Dirac bracket with every coordinate.

(b) At $k = (1,2,2)$, display the projector and project $(3,-1,2)$. Recover the annulus smeared bracket of Lecture 12 from this matrix symbol, using real vector smearings f, g.

(c) At $k = 0$, which step fails? Explain why periodic real gauge parameters on $\mathbb{T}^3$ leave three constant canonical pairs, and give their form.

Exercise 12.2 (A union of annuli, rather than one annulus). For $n \geq 2$ let $\mathcal{U}_n = \{n^{-1} < |k| < n\}$ and let $\mathcal{Y}$ be the union of the real Schwartz classes with smooth compact Fourier support in $\mathcal{U}_n$.

(a) Prove that $\mathcal{Y}$ is dense in the real spaces $L^2(\mathbb{R}^3;\mathbb{R})$ and $H^1(\mathbb{R}^3;\mathbb{R})$, using the appropriate norm in each case. You may use the usual L^2 density of compact smooth functions, proved by truncation and smoothing.

(b) Prove the corresponding density for transverse vector fields, with approximants $P_{\mathrm{T}}v_n$. Keep the real-field Fourier condition.

(c) Show that the analogous class in one fixed annulus is not dense in either whole space. Explain why a bounded extension from a dense union must not be justified by density of just one annulus.

Exercise 12.3 (A gauge parameter outside L^2). Choose a real radial $\eta \in C_c^\infty(\mathbb{R}^3)$ equal to one on $|k| \le 1$. Define a tempered scalar distribution and a vector by their Fourier profiles

$$\widehat{\chi}_0(k) = \frac{\eta(k)}{|k|^2}, \qquad \widehat{A}(k) = ik\,\widehat{\chi}_0(k) \quad (k \neq 0).$$

A locally integrable function of at most polynomial growth defines a tempered distribution. You may also use that a distribution with all first derivatives zero is constant.

(a) Show that χ_0 is well-defined and real, that $A \in H^1$, and that $\chi_0 \notin L^2$. Can any L^2 scalar have gradient A?

(b) For a general real $A \in H^1(\mathbb{R}^3;\mathbb{R}^3)$ compare the two profiles $\widehat{\chi}_L = -i(k\cdot\widehat{A})/|k|^2$ and $-\widehat{\chi}_L$. Which one has gradient $(I - P_{\mathrm{T}})A$, and which gauges A to $P_{\mathrm{T}}A$? Justify that χ_L is tempered even near zero frequency.

Exercise 12.4 (Weak symplectic does not mean every gradient exists). On the real space $\mathcal{E}_T = H^1_{\mathrm{trans}} \times L^2_{\mathrm{trans}}$, use the canonical form from Lecture 12. Let $\theta(r)$ be smooth, zero for $r \le 1$ and one for $r \ge 2$, and define the real transverse L^2 vector h by

$$\widehat{h}(k) = \theta(|k|)|k|^{-2}P_{\mathrm{T}}(k)e_1.$$

(a) Prove that $h \in L^2$ but $h \notin H^1$. Show that $F(a,p) = \int h \cdot p\,\mathrm{d}^3x$ is continuous on $\mathcal{E}_T$.

(b) Show that no $X \in \mathcal{E}_T$ satisfies $\iota_X\Omega = DF$. Why does this not contradict the nondegeneracy proved in Lecture 12?

(c) Does finiteness of $\int(|\operatorname{curl} A|^2 + |\Pi|^2)$ imply membership in this inhomogeneous phase space? Test a nonzero constant potential with zero momentum.

Classical Field Theory — Exercise Sheet 13

Radiation estimates and boundary conditions

Questions only

Use Lecture 13's real Maxwell fields, retarded power theorem and amplitude $\mathcal{A}(u,n) = (4\pi)^{-1}\int \mathbf{J}(u+n\cdot y, y)\mathrm{d}^3y$. Here $n\in\mathbb{S}^2$, $u=t-r$, and $P_{\mathrm{T}}(n)=I-nn^{\mathsf{T}}$. The third problem also uses Lecture 6's time-reflected advanced kernel. The singular-source formula in the last problem is supplied separately.

Exercise 13.1 (A general slow-time source). Let $\rho_0(s,y)$ be scalar and $J_0(s,y)$ vector-valued, both real and smooth with compact spacetime support, satisfying $\partial_s\rho_0 + \mathrm{div}_y J_0 = 0$. For $0<\varepsilon\le 1$ put

$$\rho_\varepsilon(t,y)=\rho_0(\varepsilon t,y),\qquad \mathbf{J}_\varepsilon(t,y)=\varepsilon J_0(\varepsilon t,y),\qquad p_0(s)=\int y\rho_0(s,y)\mathrm{d}^3y.$$

Define finite constants

$$M_0=\int \sup_s|\partial_s J_0(s,y)|\mathrm{d}^3y,$$
$$M_1=\int |y|\sup_s|\partial_s^2 J_0(s,y)|\mathrm{d}^3y,$$
$$M_2=\frac{1}{2}\int |y|^2\sup_s|\partial_s^3 J_0(s,y)|\mathrm{d}^3y.$$

(a) Verify conservation and prove $p_0'(s)=\int J_0(s,y)\mathrm{d}^3y$. Find the scaled dipole $p_\varepsilon(t)=\int y\rho_\varepsilon(t,y)\mathrm{d}^3y$ and its second derivative.

(b) Prove, uniformly for all $u\in\mathbb{R}$, the explicit absolute estimate

$$\left|P_\varepsilon(u)-\frac{|\ddot p_\varepsilon(u)|^2}{6\pi}\right|\le\frac{2M_0M_1+M_1^2}{4\pi}\varepsilon^5.$$

(c) Improve the integrated error to $O(\varepsilon^6)$ by retaining the first delayed-sampling correction. Give an explicit bound using M_0,M_1,M_2. Why does this improvement not assert that the directional power density has an $O(\varepsilon^6)$ error?

Exercise 13.2 (A nonradial source at zero acceleration). Choose a nonnegative $f\in C_c^\infty(B_1(0))$ with $\int f=1$ and nonzero first moment $\mu=\int yf(y)\mathrm{d}^3y$. Let $a\in\mathbb{R}^3\setminus\{0\}$ and choose a smooth compactly supported p_0 equal to as^3 on $|s|\le 1$. Use the conserved dipole source

$$p(t)=p_0(\varepsilon t),\qquad \rho=-p(t)\cdot\nabla f,\qquad \mathbf{J}=\dot p(t)f,\qquad 0<\varepsilon\le 1.$$

Compute the exact retarded power $P(0)$. Show that it is positive although $\ddot p(0)=0$, and evaluate it for $a=e_3$, $\mu=e_3/4$. Explain what this rules out about a relative error estimate. You may use the angular moments

$$\int n_in_j\mathrm{d}\Omega=\frac{4\pi}{3}\delta_{ij},\qquad \int n_in_jn_kn_l\mathrm{d}\Omega=\frac{4\pi}{15}(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}).$$

Exercise 13.3 (The half-retarded, half-advanced field). For a real smooth compact spacetime conserved source, let $(\phi_{\mathrm{em}}^{\mathrm{ret}}, \mathbf{A}_{\mathrm{ret}})$ be its retarded potentials. Define the advanced potentials by replacing $t - |x-y|$ by $t + |x-y|$ in the same $1/(4\pi|x-y|)$ integrals. Time reflection of Lecture 6's kernel gives these smooth sourced solutions. Let the symmetric potentials be half the sum of the two, and form their Maxwell fields.

(a) At future null infinity, $t = r + u$, show that the advanced field is exactly zero for all sufficiently large r (at each fixed u). Find the symmetric field's limiting outward power in terms of the retarded power.

(b) At past null infinity, $t = v - r$, define

$$\overline{\mathcal{A}}(v,n) = \frac{1}{4\pi}\int \mathbf{J}(v - n\cdot y, y)\mathrm{d}^3y, \qquad \overline{\mathcal{V}}(v,n) = \frac{1}{4\pi}\int \rho(v - n\cdot y, y)\mathrm{d}^3y.$$

Derive the leading advanced electric and magnetic coefficients and the signed outward power of the symmetric field. Primes now mean ∂_v. Does averaging the potentials cancel radiation at future null infinity?

Exercise 13.4 (Liénard–Wiechert: a supplied singular-source formula). Let $z\colon \mathbb{R} \to \mathbb{R}^3$ be a smooth complete worldline with $|\dot{z}(s)| \le v_* < 1$. Fix an event (t,x) with $x \neq z(t)$. Let $Q \in \mathbb{R}$ be its charge. For this problem only, take as given the point-source potential prescription

$$\phi_{\mathrm{em}}(t,x) = \frac{Q}{4\pi}\int \frac{\delta(t - s - |x - z(s)|)}{|x - z(s)|}\mathrm{d}s, \qquad \mathbf{A}(t,x) = \frac{Q}{4\pi}\int \frac{\dot{z}(s)\delta(t - s - |x - z(s)|)}{|x - z(s)|}\mathrm{d}s.$$

The delta integrals are understood locally at their roots. You may use the one-variable identity $\delta(g(s)) = \sum_j \delta(s - s_j)/|g'(s_j)|$ for simple roots.

(a) Prove existence and uniqueness of the retarded root $t = s_{\mathrm{ret}} + |x - z(s_{\mathrm{ret}})|$. Evaluate the two potentials in terms of $R = |x - z(s_{\mathrm{ret}})|$, $n = (x - z(s_{\mathrm{ret}}))/R$ and $v = \dot{z}(s_{\mathrm{ret}})$.

(b) For $z(s) = (3s/5, 0, 0)$ at $(t,x) = (1,0)$, compute the retarded time, Jacobian and scalar potential. Explain exactly which smooth-source hypotheses fail and why this calculation establishes no self-force law.

Classical Field Theory — Exercise Sheet 14

First-order systems and real structures

Questions only

All spinor entries commute. Use Lecture 14's Hermitian inner product, unitary Fourier convention and real form $\Omega(h,g) = 2\,\mathrm{Im}\int h^\dagger g$. Problems 1 and 3 supply new four-component matrices; Problem 4 returns to the actual two-component matrices of Lecture 14.

Exercise 14.1 (Four components in three space dimensions). Let I_2 be the two-by-two identity and put

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \Gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \Gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}.$$

Define $\beta_D = \Gamma^0$, $\alpha_D^i = \Gamma^0\Gamma^i$ and $H_D(k) = \alpha_D^i k_i + m\beta_D$, where $m \geq 0$.

(a) Verify every Clifford anticommutator for signature $(+,-,-,-)$. Prove that $H_D(k)$ is Hermitian and find its eigenvalues, multiplicities and orthogonal spectral projectors, including the exceptional $m = k = 0$ case.

(b) For $\mathrm{i}\psi_t = (-\mathrm{i}\alpha_D^i\partial_i + m\beta_D)\psi$, derive the conserved current $\rho = \psi^\dagger\psi$, $j^i = \psi^\dagger\alpha_D^i\psi$ and prove $|n\cdot j| \leq \rho$ for every unit vector n. Explain why verifying these matrices alone does not prove the spin representation or Lorentz-covariance theorem cited in Lecture 14.

Exercise 14.2 (A constant symmetric-hyperbolic system). Let $A^1,\dots,A^d$ be constant Hermitian N-by-N matrices and B a constant skew-Hermitian matrix. For complex columns consider

$$w_t + A^i\partial_i w + Bw = 0, \qquad c_* = \sup_{|n|=1} \|n_i A^i\|_{\mathrm{op}}.$$

Here the operator norm is the Euclidean matrix norm.

(a) Construct the unique distributional solution in $C(\mathbb{R};L^2)$ for each L^2 datum and prove norm preservation. Derive the smooth local current.

(b) Using Lecture 5's moving-ball argument, prove that data supported in a closed set K remain supported in $\{x : \mathrm{dist}(x,K) \leq c_*|t|\}$. Give the limiting argument for rough data and include $c_* = 0$.

(c) For $d = N = 2$, take $A^1 = \sigma_1$, $A^2 = 2\sigma_3$ and $B = \mathrm{i}\sigma_2$. Compute c_* and exhibit a column and normal attaining the current bound. Would Hermitian $B = I_2$ preserve the same norm?

Exercise 14.3 (Weyl sectors and a mass coupling). Use the four-component matrices of Problem 1. Define $\Gamma^5 = \mathrm{i}\Gamma^0\Gamma^1\Gamma^2\Gamma^3$ and $P_{\mathrm{ch},\pm} = (I_4 \pm \Gamma^5)/2$. Compute Γ^5 and its two eigenspaces. Write the Dirac Hamiltonian in the coordinates

$$\psi = \frac{1}{\sqrt{2}}\begin{pmatrix} u+v \\ u-v \end{pmatrix}, \qquad u,v \in \mathbb{C}^2.$$

Determine when the two chiral subspaces are invariant under evolution, and derive the two Weyl equations in the massless case.

Exercise 14.4 (A commuting Majorana-type real subspace). Return to Lecture 14's two-component matrices $\alpha = -\sigma_3$, $\beta = \sigma_1$ and define the real-linear, antilinear-in-complex-scalars map

$$C\psi = \sigma_3 \overline{\psi}.$$

Prove $C^2 = I$, characterize its fixed space in L^2, and show that Dirac evolution preserves it. Derive the evolution of its two real entries. Restrict the real form Ω to that fixed space. Is the result nondegenerate? Explain why this calculation does not give a classical commuting replacement for a quantum Majorana field or its statistics.

Classical Field Theory — Exercise Sheet 15

Beyond the one-dimensional cubic proof

Questions only

Use the mass-one Klein–Gordon group $U_d(t)$ on $\mathcal{E}_d = H^1(\mathbb{R}^d) \times L^2(\mathbb{R}^d)$, with Hilbert norm $\|(f,g)\|_{\mathrm{lin}}^2 = \|f\|_{H^1}^2 + \|g\|_2^2$. Lecture 15 proves the forced estimate in one dimension; Problem 2 checks its dimension-independent Fourier proof. All fields below are real.

Exercise 15.1 (A general potential and an affine background). Let $V \in C^3(\mathbb{R})$ and let b be a smooth bounded solution of $-b'' + V'(b) = 0$ on $\mathbb{R}$. Do not assume $b \in L^2$. For $\phi = b + v$ consider $\phi_{tt} - \phi_{xx} + V'(\phi) = 0$.

(a) Derive the mass-one perturbation force $N_b(v)$, and show that it maps $H^1(\mathbb{R})$ locally Lipschitzly to $L^2(\mathbb{R})$, vanishing at zero. Give a constant on a ball of radius R using the maximum of $|V''|$ on a specified compact interval.

(b) For every initial ball $\|(v_0, g_0)\|_{\mathrm{lin}} \le B$, $B \ge 1$, give a common contraction radius and positive time. Prove existence, uniqueness among weak energy solutions, continuous dependence, and the finite-lifespan alternative. Which derivatives of V did the estimates use?

Exercise 15.2 (The cubic survives in three dimensions). Take as given the Sobolev estimate $\|u\|_{L^6(\mathbb{R}^3)} \le C_S \|u\|_{H^1(\mathbb{R}^3)}$ for every real $u \in H^1$, where C_S is a fixed positive constant. Consider

$$u_{tt} - \Delta u + u^3 = 0, \qquad (u(0), u_t(0)) \in \mathcal{E}_3.$$

(a) Justify the three-dimensional version of the forced mass-one energy estimate, including the weak equation. Prove local existence, energy-class uniqueness and continuous dependence by a Duhamel contraction, with a common lifespan for a bounded initial ball.

(b) Would the same $H^1 \to L^2$ force argument handle u^5? For a nonzero real $\eta \in C_c^\infty(\mathbb{R}^3)$, compute the scaling of $u_\Lambda(x) = \Lambda^{1/2}\eta(\Lambda x)$, $\Lambda \ge 1$, in H^1, and of its cubic and quintic forces in L^2. Limit the conclusion to this contraction argument.

Exercise 15.3 (Sine-Gordon on a weighted path space). Let b be a smooth bounded stationary sine-Gordon background, $-b'' + \sin b = 0$, and write $\phi = b + v$. For any fixed $T > 0$, equip $X_T = C([0,T]; \mathcal{E}_1)$ with the equivalent norm

$$\|z\|_a = \sup_{0 \le t \le T} e^{-at} \|z(t)\|_{\mathrm{lin}}, \qquad a > 0.$$

Derive the reference mass-one Duhamel force. Show that the Duhamel map is a contraction on the entire space X_T for $a = 4$, with no radius restriction. Prove existence and uniqueness on every finite interval and give a continuous-dependence bound. Why is an amplitude-dependent local constant unnecessary? Do not assume that b or $\sin b$ belongs to L^2.

Exercise 15.4 (Defocusing continuation with the analytic input stated). For the three-dimensional cubic solution of Problem 2, assume additionally that the following energy identity holds throughout its maximal energy-class lifespan:

$$E_0 = \int_{\mathbb{R}^3} \left[\frac{1}{2} u_t^2 + \frac{1}{2} |\nabla u|^2 + \frac{1}{4} u^4 \right] \mathrm{d}^3 x.$$

This is an additional hypothesis at the weak energy regularity, not a consequence of L3's classical balance. Prove that the initial integral is finite and that the solution is global. Give a bound for its full $\mathcal{E}_3$ norm on every finite time interval, explicitly recovering the L^2 displacement term.

Classical Field Theory — Exercise Sheet 16

Energy, plateaux and local blow-up

Questions only

Use the general one-dimensional local theorem proved on Sheet 15: for $V \in C^3$, $V'(0) = 0$, data in $H^1(\mathbb{R}) \times L^2(\mathbb{R})$ have a unique weak energy solution, and a finite maximal endpoint forces its full energy-space norm to tend to infinity. Its reference linear mass is one. The conservation argument below must be made at that weak regularity.

Exercise 16.1 (A nonnegative potential beyond the quartic). Let $V \in C^3(\mathbb{R})$ satisfy $V \geq 0$ and $V(0) = V'(0) = 0$. For the energy solution of $u_{tt} - u_{xx} + V'(u) = 0$ with data (u_0, g_0), set

$$E(t) = \int_{\mathbb{R}} \left[\frac{1}{2} u_t^2 + \frac{1}{2} u_x^2 + V(u) \right] \mathrm{d}x.$$

(a) Prove finiteness, continuity and conservation of E by smoothing the actual solution in space. Justify the potential chain rule and the endpoint limits. Do not assume a density theorem for nonlinear solutions.

(b) Prove global existence and give a finite-time bound on the full $H^1 \times L^2$ norm. For $V(s) = s^2/2 + s^6/6$, obtain a stronger bound uniform in time. Identify which part of the proof needs $V \geq 0$.

Exercise 16.2 (Sine-Gordon energy does not bound every phase norm). Take $V(u) = 1 - \cos u$ and background zero.

(a) Use Problem 1 to prove energy conservation and global energy-class evolution. Verify explicitly the initial potential integrability and continuity needed for that argument.

(b) Choose a fixed nondecreasing $\eta \in C^\infty(\mathbb{R})$, zero for $s \leq 0$ and one for $s \geq 1$. For $R \geq 1$ take initial displacement

$$u_R(x) = 2\pi\eta(x + R + 1)\eta(R + 1 - x)$$

and zero velocity. Compute its energy in terms of two integrals on $[0, 1]$ and show that it is independent of R, while $\|u_R\|_2 \to \infty$. Does this contradict global existence or the finite-time bound of Problem 1?

Exercise 16.3 (Focusing blow-up from compact data on the line). Change the potential to $V(u) = -u^4/4$ on $\mathbb{R}$, so $u_{tt} - u_{xx} = u^3$. Fix $0 < T < R$ and choose a real $\chi \in C_c^\infty(\mathbb{R})$ with $0 \le \chi \le 1$ and $\chi = 1$ on $[-R, R]$. Take

$$u_0 = \frac{\sqrt{2}}{T}\chi, \qquad g_0 = \frac{\sqrt{2}}{T^2}\chi.$$

For this conditional exercise, assume additionally that the unique energy solution for the displayed data has a C^2 representative throughout its maximal energy-class lifespan $[0, T_*)$. This is a regularity hypothesis, and assumes no global existence.

(a) Compute its finite initial focusing energy. Compare on the cone $|x| \le R - t$ with $q(t) = \sqrt{2}/(T - t)$ for every common time $0 \le t < \min\{T, T_*\}$. Prove the needed local uniqueness rather than assuming a global finite-energy theorem for q.

(b) Deduce $T_* \le T$. State carefully what was proved if the maximal solution terminates earlier than T. Why do these data improve on simply placing the spatially constant ODE on all of $\mathbb{R}$?

Classical Field Theory — Exercise Sheet 17

Forced and accidental massless directions

Questions only

Use the real scalar kinetic coefficient $1/2$ and the orbit–Hessian lemma of Lecture 17. Counts of massless modes below refer to the linearised field. A vacuum background need not belong to L^2; perturbation data may be Schwartz or in the earlier energy class.

Exercise 17.1 (The full orthogonal orbit). For $N \geq 3$, $\lambda, v > 0$, consider

$$V(u) = \frac{\lambda}{8}(|u|^2 - v^2)^2, \qquad u \in \mathbb{R}^N, \qquad u_0 = vn, \quad |n| = 1,$$

with the usual $O(N)$ action and canonical kinetic term. Find the vacuum set, the stabiliser at u_0, its dimension, and the orbit tangent. Derive the Hessian and diagonalise the linearised field using the radial and tangential projectors, without first rotating n to a coordinate axis. Identify which masses are symmetry-forced. Finally discuss $N = 1$ separately: does its broken discrete symmetry force a massless mode?

Exercise 17.2 (A flat singlet still has a nonlinear restoring force). Add a real scalar z with its own canonical kinetic term and potential

$$\widetilde{V}(u, z) = \frac{\lambda}{8}(|u|^2 - v^2)^2 + z^4.$$

Let $O(N)$ act on u and trivially on z. Find the vacuum set, the Hessian kernel at $(vn, 0)$, and the subspace forced by the symmetry. Can the extra singlet direction be tangent to a curve of vacua? On the spatial torus, set $u = vn$ and take a homogeneous singlet $z(0) = A > 0$, $z_t(0) = 0$. Derive its nonlinear period as a definite integral and its dependence on A. Explain how a zero linearised mass is compatible with this nonlinear oscillation. These homogeneous data are not claimed to be L^2 data on $\mathbb{R}^d$.

Exercise 17.3 (An ineffective group factor belongs to the stabiliser). Take $G = SO(3)\times U(1)$ acting on a real triplet by $(R, e^{i\theta})u = Ru$. Use the radial potential of Problem 1 with $N = 3$ and vacuum $u_0 = v(2, -1, 2)/3$. Determine the full stabiliser and the differential of the orbit map $(c, \theta) \mapsto c \times u_0$ on $\mathbb{R}^3 \oplus \mathbb{R}$. Find a basis of its kernel, the number of broken directions and the Hessian nullity. Why would counting $\dim G$ or only counting rotations that fix u_0 give a wrong answer?

Exercise 17.4 (Kinetic normalisation before diagonalising masses). For a real three-component perturbation h, consider the quadratic density

$$L = \frac{1}{2}\partial_\mu h^{\mathsf{T}} K \partial^\mu h - \frac{1}{2} h^{\mathsf{T}} M h, \qquad K = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 9 & 9 \\ 0 & 9 & 9 \end{pmatrix}.$$

The real phase data may be taken Schwartz. Verify the factorisation $K = S^{\mathsf{T}} S$ for

$$S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Derive the canonical momentum, and find the corresponding change of coordinates and momenta that preserves $\int \delta h_a \wedge \delta p_a$. Compute the physical squared masses and compare them with the raw eigenvalues of M. State the general eigenproblem and explain what happens to the kernel dimension under this kinetic normalisation.

Classical Field Theory — Exercise Sheet 18

Spectrum and moving kinks

Questions only

Retain $b(x) = \tanh x$, $V(s) = \frac{1}{2}(1-s^2)^2$ and $H_K = -\partial_x^2 + 4 - 6\operatorname{sech}^2 x$ with domain $H^2 \subset L^2$. Lecture 18 proves $H_K = A^\dagger A$ for $A = \partial_x + 2\tanh x$ and $\ker H_K = \operatorname{span}\{\operatorname{sech}^2 x\}$. Use its H^1 form $\mathfrak{q}[h] = \|Ah\|_2^2$ and density/integration-by-parts results.

Exercise 18.1 (The full spectrum through two adjoint pairs). The following analytic spectral theorem is given for this problem: if W is real, bounded and smooth, with $W(x) \to 0$ at both ends, then $-\partial_x^2 + 4 + W$ on $H^2(\mathbb{R})$ is self-adjoint, its essential spectrum is $[4, \infty)$, and every spectral point outside that interval is an isolated finite-multiplicity eigenvalue. This follows from Teschl, *Mathematical Methods in Quantum Mechanics* (2009), Lemmas 9.33 and 9.35, pp. 215–216, after shifting by four; the discrete/essential spectrum convention is in §6.4, p. 145. This input supplies more than the formal factorisation in Lecture 18. Set $B = \partial_x + \tanh x$ with domain H^1.

(a) Find $B^\dagger$ and the product domains, and prove $AA^\dagger = B^\dagger B + 3$ and $BB^\dagger + 3 = -\partial_x^2 + 4$. Use these identities to classify all L^2 eigenvalues of H_K, including possible embedded or threshold eigenvalues. Check the domains when intertwining.

(b) Normalise its two bound states. Exhibit a bounded solution at spectral value four and explain why it is not another L^2 eigenstate. Using the given theorem, state the full spectrum.

Exercise 18.2 (A moving kink and the sign of momentum). For $|v_{\mathrm{tr}}| < 1$, set $\gamma = (1 - v_{\mathrm{tr}}^2)^{-1/2}$ and

$$\phi(t, x) = \tanh\bigl(\gamma(x - v_{\mathrm{tr}} t - x_0)\bigr).$$

Verify the nonlinear field equation directly. At fixed time compute the charge, full energy and the lower-index translation integral $P_1 = \int \phi_t \phi_x \mathrm{d}x$. Find the physical raised momentum P^1 and the relation between its square and the energy square. Check the result at $v_{\mathrm{tr}} = 3/5$. Justify membership in the affine class $\phi - b \in H^1$, $\phi_t \in L^2$.

Exercise 18.3 (Removing the linear translation drift). Use the spectral theorem for the self-adjoint operator in Problem 1 as an additional analytic input: on the orthogonal complement of its zero mode, the form-domain inequality is $\mathfrak{q}[h] \geq 3\|h\|_2^2$ for $h \in H^1$ with $(\psi_0, h) = 0$. Prove $\|h\|_{H^1}^2 \leq 2\mathfrak{q}[h]$ on this subspace. For any real solution of $h_{tt} + H_K h = 0$ with

$$h \in C(H^2) \cap C^1(H^1) \cap C^2(L^2),$$

prove that $(\psi_0, h(0)) = (\psi_0, h_t(0)) = 0$ preserves this orthogonality, and deduce a uniform bound on $\|h(t)\|_{H^1}^2 + \|h_t(t)\|_2^2$ from its initial quadratic energy. No existence theorem for this new linear equation is required: the assertion is conditional on a solution in the displayed class.

Exercise 18.4 (The nonlinear energy remainder on a fixed slice). For real $h \in H^1(\mathbb{R})$, compute the exact remainder in $E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] - \mathfrak{q}[h]/2$. Prove a bound using only $\|h\|_{H^1}$. If $(\psi_0, h) = 0$, show that for $\|h\|_{H^1} \le 1/(16\sqrt{2})$,

$$E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] \ge \frac{1}{8}\|h\|_{H^1}^2.$$

Why is this static estimate insufficient by itself to prove nonlinear orbital stability of the kink?

Classical Field Theory — Exercise Sheet 19

Adjoint identities, matter and surface currents

Questions only

All fields and coefficients on this sheet are smooth. Variations and gauge parameters in action identities have compact support inside the coordinate region. Use the Euler sign $\delta S = \int E_a \delta\phi^a$ and the connection conventions of Lecture 19. No equations of motion are imposed unless explicitly stated.

Exercise 19.1 (A second-order adjoint and the elementary converse). On an open coordinate patch $U \subset \mathbb{R}^{1+1}$ with coordinates (t, x), let a scalar parameter $\varepsilon \in C_c^\infty(U)$ produce two variations

$$R(\varepsilon)^1 = (1+tx)\varepsilon_{tt} + 2(t+x^2)\varepsilon_{tx} + e^t\varepsilon_x, \qquad R(\varepsilon)^2 = (1+t^2)\varepsilon_{xx} + x\varepsilon.$$

Compute $R^*(E_1, E_2)$, first in divergence form and then with every coefficient derivative expanded. Prove the compact pairing identity. Suppose an action has the compact first variation $\int E_a[\phi]h^a$ and $R_\phi^* E[\phi] = 0$ on U for every smooth field. Prove that $R_\phi(\varepsilon)$ is an infinitesimal gauge symmetry. Explain why the same converse holds for any finite-order operator $R_\phi(\varepsilon)^a = \sum_I r_\alpha^{aI}[\phi]\partial_I\varepsilon^\alpha$, without interpreting it as a converse for arbitrary conserved currents.

Exercise 19.2 (The Euler term carried by charged matter). Write $A_{\mathrm{em}} = A_\mu \mathrm{d}x^\mu$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and let $u = (u_1, u_2)^{\mathsf{T}}$ be two real scalar fields. Given $q, \lambda, v > 0$, set

$$T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D_\mu u = \partial_\mu u - qA_\mu T u, \qquad L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}D_\mu u \cdot D^\mu u - \frac{\lambda}{8}(|u|^2 - v^2)^2.$$

The gauge variations are $\delta A_\mu = \partial_\mu \varepsilon$ and $\delta u = q\varepsilon T u$.

(a) Verify covariance of $D_\mu u$, derive both Euler arrays by compact variation, and derive the off-shell Noether identity containing them.

(b) On a coordinate patch take $A_\mu = 0$ and $u = (t, t^2)^{\mathsf{T}}$. Compute the ordinary divergence of the total gauge Euler array and its balancing scalar Euler term. Does the pure Maxwell identity remain an identity for that total array by itself?

(c) For a real orthogonal representation with skew generators t_a, use instead the supplied variations $\delta A_\mu = D_\mu \varepsilon$, $\delta u = -\varepsilon^a t_a u$. Assume these variations are a gauge symmetry for every compact parameter. If the compact first variation is $\int [\langle E_A^\mu, \delta A_\mu\rangle + E_u \cdot \delta u]$, derive its covariant component identity. Here A is a connection as in Lecture 19, and the Lie algebra basis is orthonormal in its invariant pairing.

Exercise 19.3 (Noncommuting constant connections). Let $T_a = -\mathrm{i}\sigma_a/2$, so $[T_a, T_b] = \epsilon_{abc}T_c$. In pure Yang–Mills theory with coupling $g > 0$, take first $A_1 = T_1$, $A_2 = T_2$, $A_0 = A_3 = 0$, all constant. Compute every nonzero curvature and Euler component $E^\nu = g^{-2}D_\mu F^{\mu\nu}$, and check the covariant identity. Repeat after changing A_3 to $T_1 + T_2$. Exhibit the separate nonzero commutator contributions to $D_\nu E^\nu$ in that second case. What can these constant examples establish about an ordinary-divergence identity for arbitrary nonconstant connections?

Exercise 19.4 (A local superpotential theorem and a Maxwell verification). For this exercise, assume the following local decomposition statement. It is an additional hypothesis here, not a theorem to infer from the fundamental lemma alone. For a finite-order local Lagrangian of unconstrained $\mathbb{R}^N$-valued fields on a star-shaped coordinate patch of spacetime dimension at least two, a finite-order infinitesimal gauge symmetry has a local Noether current of the form

$$j_\varepsilon^\mu = W_\varepsilon^\mu + \partial_\nu U_\varepsilon^{\nu\mu}, \qquad U_\varepsilon^{\nu\mu} = -U_\varepsilon^{\mu\nu}.$$

The current and these expressions depend linearly on the parameter and finitely many of its derivatives, with smooth finite-order local field coefficients. The remainder W vanishes when the Euler expressions and their derivatives vanish. This is a local statement on the full jet space; no global topological superpotential theorem is supplied.

Verify it directly for

$$L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + A_\mu j_{\mathrm{ext}}^\mu, \qquad \partial_\mu j_{\mathrm{ext}}^\mu = 0, \qquad \delta A_\mu = \partial_\mu \varepsilon,$$

where the external current is fixed and smooth. Derive the gauge Noether current, its Euler remainder and an explicit U. On solutions, compare its spatial integral for compact spatial parameters with a parameter equal to a constant ε_∞ outside a ball. For the latter comparison assume a smooth Maxwell field whose external current has compact spatial support on the slice and total charge $Q = \int j_{\mathrm{ext}}^0 \mathrm{d}^3x$. Use $F_{i0} = E_i$ and an outward sphere orientation to find the surface sign. No Schwartz or L^2 condition on the potential is imposed in this boundary comparison.

Classical Field Theory — Exercise Sheet 20

Nonabelian constraints and local reduction

Questions only

Use $T_a = -\mathrm{i}\sigma_a/2$ for $\mathfrak{su}(2)$, so $[T_a, T_b] = \epsilon_{abc}T_c$ and $\langle T_a, T_b\rangle = \delta_{ab}$. All spatial fields are smooth on $\mathbb{T}^3$. Write $\mathcal{V}_{\mathrm{per}} = (2\pi)^3$ for its volume. The canonical bracket has the order fixed in Lectures 7 and 20.

Exercise 20.1 (A homogeneous constraint manifold). Restrict the spatial fields to constants $A_i = a_i^a T_a$ and $\Pi_i = p_i^a T_a$. A parameter $\varepsilon = \varepsilon^a T_a$ is also constant.

(a) Pull the canonical symplectic form back to these constants and derive their finite-dimensional bracket. Compute the generator $C[\varepsilon] = \mathcal{V}_{\mathrm{per}} \sum_i \langle \Pi_i, [A_i, \varepsilon]\rangle$ and its Hamiltonian vector field. Prove its constraint bracket, including the volume factor and sign.

(b) At $A_i = T_i$ for $i = 1, 2, 3$, write $\Pi_i = \sum_j M_{ij} T_j$ for a real matrix M. Classify all M satisfying $\sum_i [A_i, \Pi_i] = 0$. Give one nonscalar matrix in the constraint set and one outside it. Explain why checking a single homogeneous state cannot prove the constraint algebra on the full periodic field space.

Exercise 20.2 (Integrating a residual gauge generator). For a fixed smooth periodic $\varepsilon(x)$, let $h_s(x) = \exp(s\varepsilon(x))$ and define

$$A_i(s) = h_s^{-1} A_i h_s + h_s^{-1}\partial_i h_s, \qquad \Pi_i(s) = h_s^{-1}\Pi_i h_s.$$

Here s is the transformation parameter, not physical time.

(a) Prove that these arrays solve $\partial_s A_i(s) = D_i^{(s)}\varepsilon$ and $\partial_s \Pi_i(s) = [\Pi_i(s), \varepsilon]$. Show that the transformation preserves the canonical form, the temporal-gauge Hamiltonian and the Gauss constraint surface.

(b) Take $\varepsilon = \sin x^1\, T_3$, $A_1 = T_1$, $A_2 = T_3$, $A_3 = 0$, $\Pi_1 = T_2$ and $\Pi_2 = \Pi_3 = 0$. Compute the transformed fields and their Gauss array. Is this example on the constraint surface? What changes if a finite gauge transformation depends on physical time?

Exercise 20.3 (A linearized reduced bracket and its limit). Fix a real constant η and the background $A_1^0 = \eta T_3$, $A_2^0 = A_3^0 = 0$, $\Pi^0 = 0$. Work only with smooth variations a_i, p_i and parameters having zero spatial mean. This excludes the entire zero Fourier mode from this problem. Put $D_i^0 = \partial_i + [A_i^0, \cdot\,]$.

(a) Derive the linearized Gauss equation $D_i^0 p_i = 0$ and the gauge variation $a_i \mapsto a_i + D_i^0 \varepsilon$. For the Coulomb condition $\partial_i a_i = 0$, compute the eigenvalues of $M = -\partial_i D_i^0$ on each nonzero Fourier mode, using $T_+ = T_1 + \mathrm{i}T_2$, $T_- = T_1 - \mathrm{i}T_2$ and T_3.

(b) For $|\eta| < 1$, prove that M^{-1} maps smooth zero-mean arrays to themselves. Define

$$P = 1 + D^0 M^{-1}\,\mathrm{div}, \qquad P^* = 1 + \nabla M^{-1} D^0 \cdot\,.$$

Prove that Pa is divergence-free and gauge invariant, while $P^* p$ satisfies linearized Gauss. Prove $P^2 = P$. For real smooth zero-mean vector smearings f, g, compute the canonical bracket of $\mathcal{A}[f] = \int \langle f, Pa\rangle$ and $\mathcal{P}[g] = \int \langle g, P^* p\rangle$.

(c) At $\eta = 2$, find a real zero-mean parameter with $M\varepsilon = 0$ but $D^0\varepsilon \neq 0$. Explain why this prevents using the preceding inverse there. Does the calculation establish a global nonlinear reduced bracket?

Classical Field Theory — Exercise Sheet 21

Vector mass matrices and longitudinal energy

Questions only

Use the signature $(+,-,-,-)$ and canonically normalised real scalar kinetic coefficient $1/2$. The questions concern quadratic theories near a fixed vacuum. They make no assertion about a global unitary gauge.

Exercise 21.1 (An orbit determines a vector mass matrix). Let a compact Lie group act orthogonally on $\mathbb{R}^N$, and choose a basis of its Lie algebra, represented by skew real generators T_a, $1 \le a \le r$. Let u_0 be a vacuum of an invariant C^2 potential. Fix $g > 0$, and give the scalar the kinetic term

$$\frac{1}{2}(D_\mu\phi)\cdot(D^\mu\phi), \qquad D_\mu\phi = \partial_\mu\phi + gA^a_\mu T_a\phi.$$

The vector kinetic terms are canonical at the quadratic level.

(a) Derive the vector squared-mass matrix at $\phi = u_0$. Prove that it is nonnegative and identify its kernel and rank. Include generators which act trivially in the representation.

(b) For the three-dimensional rotation representation, take

$$T_a w = e_a \times w, \qquad u_0 = (v/\sqrt{2}, v/\sqrt{2}, 0), \qquad v > 0.$$

Compute the full matrix, its orthonormal eigenvectors and the vector masses. Which gauge-field combination remains massless?

Exercise 21.2 (The charged and neutral electroweak combinations). The following bosonic representation data are given. A complex doublet $\Phi \in \mathbb{C}^2$ has kinetic term $\frac{1}{2}(D_\mu\Phi)^\dagger D^\mu\Phi$, potential $\lambda(\Phi^\dagger\Phi - v^2)^2/8$, and vacuum $\Phi_0 = (0, v)^{\mathsf{T}}$. Here $g, g', v, \lambda > 0$ and

$$D_\mu = \partial_\mu - \mathrm{i}\left(gW^a_\mu\frac{\sigma_a}{2} + g'B_\mu\frac{I_2}{2}\right).$$

All four real vector fields have canonical quadratic Maxwell kinetic terms. The Pauli matrices and the hypercharge generator $Y = I_2/2$ specify the representation; no fermion sector is included.

(a) Find the full real squared-mass matrix in the order (W^1, W^2, W^3, B). Diagonalise its neutral block and define Z and the massless photon A_γ with unit kinetic normalisation. Define $W^\pm = (W^1 \mp \mathrm{i}W^2)/\sqrt{2}$ and find their common mass.

(b) Find the generator annihilating Φ_0 and the coefficient of its photon coupling in D_μ. Find the radial scalar mass and compare the quadratic number of physical modes in the uncoupled limit $g, g' \to 0$ with the broken quadratic description at positive couplings. Explain why W^+ and W^- do not count as two independent complex vectors.

This is the bosonic mass calculation of the electroweak model. For the original construction see Weinberg, *Physical Review Letters* **19** (1967), 1264–1266, equations (4), (7)–(12). The signs and scalar normalisation to use are those explicitly given above.

Exercise 21.3 (The positive reduced Proca Hamiltonian). Use Lecture 21's variables $B = -\beta\,\mathrm{d}t + \mathbf{b}\cdot\mathrm{d}x$, mass $m > 0$ and spatial momentum $\mathbf{\Pi}$. Take real Schwartz data on $\mathbb{R}^3$. Before elimination the Hamiltonian is

$$H_B = \int_{\mathbb{R}^3} \left[\frac{1}{2}|\mathbf{\Pi}|^2 + \frac{1}{2}|\operatorname{curl}\mathbf{b}|^2 + \frac{1}{2}m^2|\mathbf{b}|^2 + \beta \operatorname{div}\mathbf{\Pi} - \frac{1}{2}m^2\beta^2 \right] \mathrm{d}^3x.$$

Its scalar constraints are $\pi_\beta = 0$ and $\operatorname{div}\mathbf{\Pi} - m^2\beta = 0$.

(a) Eliminate the scalar pair and derive the reduced Hamiltonian and both spatial Hamilton equations with the remaining canonical form. Prove positivity and show that a zero-energy real datum is zero.

(b) At a nonzero Fourier vector k, split the spatial amplitudes into longitudinal and transverse parts. Find the frequencies and a canonical normalisation of each real longitudinal Fourier quadrature. Treat $k = 0$ separately. Why does the energy formula not have a regular mass-zero limit for fixed arbitrary longitudinal momentum?

Individual Fourier modes here analyse the multiplier; they are not claimed to be finite-energy plane-wave fields on $\mathbb{R}^3$.

Classical Field Theory — Exercise Sheet 22

Vortex orientation, scalar reduction and moduli

Questions only

Throughout this sheet $q = v = \lambda = 1$, $D_j = \partial_j - \mathrm{i}A_j$, $B = \partial_1 A_2 - \partial_2 A_1$, and the plane has orientation $\mathrm{d}x_1 \wedge \mathrm{d}x_2$. Use Lecture 22's energy and current $K_j = \frac{1}{2}\operatorname{Im}(\overline{\Phi} D_j \Phi)$. Every sector statement retains its finite-energy, uniform vacuum-modulus and absolute magnetic-integrability assumptions.

Exercise 22.1 (Reflection and higher-winding antivortices). Let $R(x_1, x_2) = (x_1, -x_2)$. For smooth fields set $\widetilde{\Phi} = \Phi \circ R$ and $\widetilde{A} = R^* A$.

(a) Track B, K, the energy and counterclockwise winding under this reflection. Derive the square completion for winding $-N$, $N \geq 1$, and prove the bound and equality equations in the reflected sector.

(b) For real radial profiles use $\Phi = f(r)e^{-\mathrm{i}N\theta}$ and $A = \alpha(r)\mathrm{d}\theta$. Derive both equality ODEs, the flux and the boundary conditions on the positive vacuum branch. Give sufficient smooth Cartesian core forms and the first corrections to $f = s_N r^N + \cdots$ and $\alpha = -r^2/4 + \cdots$. Here $s_N > 0$ is a local slope coefficient, not a proved global shooting value.

Exercise 22.2 (The scalar equation with its point sources). Let a positive critical vortex in the stated sector have finitely many zeros $z_j \in \mathbb{R}^2$ of multiplicities $m_j \geq 1$. Away from the zeros put $w = \log|\Phi|^2$. Assume as local zero data that $w = 2m_j \log|x - z_j| + w_j$ near z_j, where w_j is smooth. This local regular remainder is a hypothesis, not a claim about arbitrary smooth fields with zeros. Use the distributional identity $\Delta \log|x - z_j| = 2\pi\delta_{z_j}$ as a given.

(a) On a phase patch derive A_1, A_2 in terms of w and the real phase. Eliminate the potential from the two Bogomolny equations.

(b) Prove the resulting equation on the whole plane as a distribution, including its point-source coefficient. Determine $\int_{\mathbb{R}^2}(1 - e^w)\mathrm{d}^2x$ from the sector flux when $\sum_j m_j = N$. Does this reduction prove existence for prescribed zeros?

Exercise 22.3 (Parameters at a two-vortex collision). A *moduli space* here is the set of critical positive N-vortex solutions with the vacuum asymptotics, modulo smooth gauge transformations. Use the following stronger geometric result as a *given*: this space is a nonsingular complex manifold, and the coefficients of

$$P(z) = \prod_{j=1}^{N} (z - z_j)$$

are smooth complex coordinates, including at coincident zeros. This input is not inferred from existence and uniqueness as a set alone. For its zero-coordinate description see Alonso Izquierdo, Garcia Fuertes, Manton and Mateos Guilarte, *JHEP* **01** (2024) 020, §5, pp. 10–11, equation (5.1) and discussion. Taubes's existence/uniqueness result is the one cited in Lecture 22.

(a) Count the real moduli and the moduli after fixing the mean zero position. Explain why ordering the zeros overcounts configurations.

(b) For $N = 2$, write $P(z) = z^2 - sz + p$. Replace (s, p) by the centre $Z = s/2$ and discriminant $\Delta = s^2 - 4p$. Locate the collision set and determine its real codimension. Explain why (z_1, z_2) and their separation are poor smooth coordinates there.

Exercise 22.4 (Designing a radial computation). For the positive unit-vortex ODEs $f' = (1 - \alpha)f/r$ and $\alpha'/r = (1 - f^2)/2$, design a computation on $[r_0, R]$, where $0 < r_0 \ll 1 < R$.

(a) Derive the coefficients through r^5 in f and through r^6 in α, using the free coefficient $s_0 = f'(0)$.

(b) Treat s_0 as an unknown parameter. Give three boundary residuals for a two-equation boundary-value solver using the core series and $f(R) = 1$. Explain why adding $\alpha(R) = 1$ independently would overdetermine this particular setup.

(c) Specify separate tests of origin truncation, outer-boundary truncation and solver accuracy. What further information is needed before a numerical residual could prove existence of a full-plane vortex? No numerical run is required in this problem.

Classical Field Theory — Exercise Sheet 23

A second KdV bracket and further conserved quantities

Questions only

Use Lecture 23's field space $\mathcal{S} = \mathcal{S}(\mathbb{R}, \mathbb{R})$, gradient space $\mathcal{B} = \mathbb{R} \oplus \mathcal{S}$ and admissible smooth functionals $\mathcal{A}_{\mathrm{KdV}}$. Write $D_x = \partial_x$ for spatial differentiation. Every time-dependent conclusion is conditional on a C^1 path in $\mathcal{S}$ satisfying KdV.

Exercise 23.1 (A variable Hamiltonian operator). For $u \in \mathcal{S}$ define the linear operator $J_2[u] : \mathcal{B} \to \mathcal{S}$ by

$$J_2[u]b = -b''' + 4ub' + 2u'b.$$

The prime means D_x, including when applied to a gradient. Define $\{F, G\}_2 = \int_{\mathbb{R}} F_u J_2[u] G_u \, \mathrm{d}x$.

(a) Prove skewness by integration by parts with both gradients in $\mathcal{B}$, keeping the endpoint primitive. Derive the gradient of $\{F, G\}_2$ and prove closure of the stated class.

(b) Prove Jacobi, also for $J_2 + \alpha D_x$ with every constant $\alpha \in \mathbb{R}$. Use the Hessian symmetry from Lecture 23 and account for the extra terms from differentiating the operator itself. A useful notation is $W(a, b) = ab' - a'b$.

(c) For $P = \frac{1}{2}\int u^2$ and $H = \int (u^3 + u_x^2/2)$, prove $J_2 P_u = D_x H_u$. Is the Gardner mass Casimir also a Casimir of this bracket?

The second KdV operator is discussed in Olver, *Applications of Lie Groups to Differential Equations*, second edition, Examples 7.6 and 7.9, pp. 438–439 and 444. The operator above states the course's scaling and signs explicitly; its proof is part of this problem.

Exercise 23.2 (Finding a higher polynomial invariant). For real constants A, B, C put

$$I_{A,B,C}[u] = \int_{\mathbb{R}} \left(Au^4 + Bu(u')^2 + C(u'')^2 \right) \mathrm{d}x.$$

Determine all triples such that $DI_{A,B,C}(u)(6uu' - u''') = 0$ for every $u \in \mathcal{S}$. Conclude conservation along every stated Schwartz KdV path. Show necessity as well as sufficiency; do not infer independence of two integrals merely by looking at different integrands.

Exercise 23.3 (Classifying Gardner Casimirs). A Casimir of Lecture 23's Gardner bracket is an admissible functional commuting with every admissible functional. Prove that these are exactly

$$F[u] = \psi(M[u]), \qquad M[u] = \int_{\mathbb{R}} u \, \mathrm{d}x,$$

for smooth $\psi : \mathbb{R} \to \mathbb{R}$. Constant scalar functionals are included. Explain why this does not say that every local polynomial density is a nonlinear function of the density u. Compare the local-functional discussion in Olver, Example 7.13, p. 446: the present admissible class also allows nonlocal composites $\psi(M)$.

Classical Field Theory — Exercise Sheet 24

A higher Lax flow and operator domains

Questions only

Use $D_x = \partial_x$ and the ordered commutator $[B, L] = BL - LB$. All operator calculations in the first problem are on $C_c^\infty(\mathbb{R}, \mathbb{C})$, with smooth real coefficient u. A displayed differential expression does not specify its Hilbert-space domain.

Exercise 24.1 (Determining a fifth-order Lax flow). Let $L = -D_x^2 + u$. For real constants $r_1, \dots, r_6$ consider

$$B_5 = 16D_x^5 + r_1 u D_x^3 + r_2 u' D_x^2 + (r_3 u'' + r_4 u^2) D_x + r_5 u''' + r_6 u u'.$$

Products with u and its derivatives here mean multiplication before the indicated derivative of the test.

(a) Determine the constants so that $[B_5, L]$ is multiplication for every smooth u. Find the resulting scalar coefficient and write the corresponding evolution equation $L_t = [B_5, L]$.

(b) Prove that the resulting B_5 is formally skew-symmetric on compact smooth tests. Distinguish this from a statement about a closed skew-adjoint operator on L^2.

Exercise 24.2 (Constancy for a sufficiently regular eigenbranch). Let $u \in C^1(I; \mathcal{S}(\mathbb{R}, \mathbb{R}))$ solve $u_t = K_5[u]$ on an open interval I. Suppose $\lambda \in C^1(I; \mathbb{R})$ and $g \in C^1(I; H^7(\mathbb{R}, \mathbb{C}))$, with $g(t) \neq 0$ and $L(t)g(t) = \lambda(t)g(t)$ at every time. The Sobolev space H^7 uses the Fourier norm $\|\langle k \rangle^7 \widehat{g}\|_2$.

Use the following operator fact as a *given*: a real bounded multiplication perturbation of $-D_x^2$ is self-adjoint on the common domain $H^2(\mathbb{R})$. This is the relative-bound-zero case of Kato, *Perturbation Theory for Linear Operators*, V, Theorem 4.3, p. 287. Prove that λ is constant. Specify the domain needed for moving L across an inner product. Does this calculation require simplicity or isolation, and does it construct a branch or prove a theorem about the whole spectrum?

Exercise 24.3 (The same interior expression with moving endpoints). Fix $\ell > 0$ and a smooth positive nonconstant function $\kappa(t)$. On $L^2(0, \ell)$ consider the expression $L = -\partial_x^2$ with domain

$$\mathcal{D}_t = \left\{ f \in H^2(0, \ell) : f(0) = 0, \quad f'(\ell) = \beta(t) f(\ell) \right\}, \qquad \beta(t) = \kappa(t) \coth(\kappa(t)\ell).$$

The trace values in this formula have their usual continuous Sobolev representatives on a finite interval.

(a) Show that this realization is symmetric by its endpoint Green formula. Find a nonzero smooth eigenfunction with a nonconstant eigenvalue. Self-adjointness is not required for this task.

(b) With $B = 0$, verify $L_t \psi = [B, L]\psi = 0$ on every compactly supported interior test. Explain why this does not contradict the preceding conditional eigenbranch result or establish constancy of the varying eigenvalue.

Classical Field Theory — Exercise Sheet 25

Analytic extension and direct scattering

Questions only

Use $L = -\partial_x^2 + u$, the Wronskian $W(f,g) = fg' - f'g$, and $f_- = af_+(\cdot,-k) + bf_+(\cdot,k)$. For real $k \neq 0$, the right reflection amplitude is $r = b/a$. All potentials in this sheet are real Schwartz functions.

Exercise 25.1 (An upper-half-plane Volterra construction). For $\operatorname{Im} k \geq 0$ define

$$H(s,k) = \int_0^s e^{2ik\sigma} d\sigma, \qquad s \geq 0.$$

Thus $H(s,k) = (e^{2iks} - 1)/(2ik)$ for $k \neq 0$ and $H(s,0) = s$. Consider

$$m(x,k) = 1 + \int_x^\infty H(y-x,k)u(y)m(y,k)dy.$$

(a) For every finite R, construct the unique bounded solution on $x \geq R$ by iteration, uniformly for k in the closed upper half-plane. Prove that $m(x,k)$ is holomorphic in $\operatorname{Im} k > 0$ for each fixed x, and continuous up to its real boundary, including $k = 0$.

(b) Prove $m \to 1$ and $m_x \to 0$ as $x \to +\infty$. Show that $f_+(x,k) = e^{ikx}m(x,k)$ solves $Lf_+ = k^2 f_+$. State the two right asymptotic conditions at zero energy. Does this prove a nonzero Wronskian basis at $k = 0$?

The complex-analysis inputs may be used as givens: dominated parameter integrals of holomorphic functions are holomorphic, and locally uniform limits of holomorphic functions are holomorphic. These are the parameter-integral theorem and the uniform-limit result proved in *Analysis*, Lecture 10.

Exercise 25.2 (Reflectionless scattering by intertwining). Fix $\kappa > 0$ and set

$$L_\kappa = -\partial_x^2 - 2\kappa^2 \operatorname{sech}^2(\kappa x), \qquad A = \partial_x + \kappa \tanh(\kappa x), \quad A^\dagger = -\partial_x + \kappa \tanh(\kappa x).$$

(a) Factor $L_\kappa + \kappa^2$ and the free operator plus κ^2. Use the resulting intertwining identity to construct both real-energy Jost solutions and determine a, b, r with the fixed Wronskian convention.

(b) Find the right-normalised negative-energy state and its inverse squared norm. Show that there are no other negative-energy L^2 states. Then give a bounded zero-energy solution which is not in L^2.

Use the Sobolev integration-by-parts and domain arguments of Lecture 18. A bounded non-L^2 zero-energy solution is called a *zero-energy resonance* in this problem.

Exercise 25.3 (A weak asymmetric potential retains a phase). Let $u_\epsilon = \epsilon V$, with $\epsilon \in \mathbb{R}$, and

$$V(x) = e^{-(x-1)^2} + 2e^{-(x+2)^2}.$$

For each fixed real $k \neq 0$, derive the first-order expansions of a, b, r as $\epsilon \to 0$ and prove an $O(\epsilon^2)$ remainder bound. Evaluate the Gaussian coefficient and the leading reflection probability. Show that no negative-energy bound states occur when $\epsilon \geq 0$. The expansion is a fixed-k assertion; no uniform limit at $k = 0$ is requested.

Classical Field Theory — Exercise Sheet 26

Scattering evolution and soliton shifts

Questions only

Use the standing real KdV solution $u \in C^\infty(\mathbb{R};\mathcal{S})$ of Lecture 24 and the Jost/scattering conventions of Lecture 25. The first problem fills the analytic step left sketched in Lecture 26. The later finite-rank construction prescribes its norming coefficients; it makes no additional theorem about birth of eigenvalues.

Exercise 26.1 (Time differentiation at real energy). Fix real $k \neq 0$. For the right Jost solution put $f_+(t,x,k) = e^{ikx}m(t,x,k)$, with

$$m = 1 + \int_x^\infty \frac{e^{2ik(y-x)} - 1}{2ik}\, u(t,y)m(t,y,k)\mathrm{d}y.$$

(a) Prove that this Volterra construction is continuously differentiable in t. On every compact time interval prove $m_t \to 0$ as $x \to +\infty$, together with the spatial derivative limits needed to compare both the function and derivative tails of $(f_+)_t - Bf_+$. Here $B = -4\partial_x^3 + 6u\partial_x + 3u_x$.

(b) Prove the right and left temporal equations, justify differentiability of a, b from finite-point Wronskians, and prove $r(k,t) = r(k,t_0)e^{8ik^3(t-t_0)}$. Do not pair the oscillatory Jost solutions in L^2.

Exercise 26.2 (A positive finite-rank inverse equation). Take distinct $\kappa_j > 0$ and $c_j > 0$, $1 \le j \le N$, with $F(s) = \sum_j c_j e^{-\kappa_j s}$. Consider the right Marchenko equation

$$K(x,y) + F(x+y) + \int_x^\infty K(x,z)F(z+y)\mathrm{d}z = 0, \qquad y \ge x.$$

The finite positive reflectionless case of the inverse-scattering theorem is a *given*: these data are admissible and reconstruct a real rapidly decaying potential by $u = -2\,\mathrm{d}K(x,x)/\mathrm{d}x$. It is the zero-background specialization of the cited theorem in Lectures 25–26: Egorova–Grunert–Teschl, *Nonlinearity* **22** (2009), §4, Theorems 4.3–4.4, equations (4.26)–(4.30). Their $(\gamma_j^+)^2$ is c_j.

(a) Reduce the equation to a real symmetric $N \times N$ linear system. Prove that its matrix is positive definite and invertible for every real x.

(b) With $v_j(x) = \sqrt{c_j}e^{-\kappa_j x}$, define $\mathsf{C}_{jl} = v_j v_l/(\kappa_j + \kappa_l)$ and $\tau = \det(I + \mathsf{C})$. Prove $K(x,x) = (\log\tau)'$ and $u = -2(\log\tau)''$. Explain why no real pole arises from these positive data.

Exercise 26.3 (Two pulses and their shifts). Take $\kappa_1 > \kappa_2 > 0$ and prescribe $c_j(t) = c_j(0)e^{8\kappa_j^3 t}$, with $c_j(0) > 0$. Put

$$x_{j,0} = \frac{1}{2\kappa_j}\log\frac{c_j(0)}{2\kappa_j}, \qquad \eta_j = -2\kappa_j(x - 4\kappa_j^2 t - x_{j,0}).$$

(a) Evaluate the two-state determinant and find its interaction coefficient d_{12} in $\tau = 1 + e^{\eta_1} + e^{\eta_2} + d_{12}e^{\eta_1+\eta_2}$.

(b) Verify directly that $u = -2\partial_x^2 \log\tau$ solves KdV. One may use, after deriving it by differentiating $\log\tau$,

$$\tau\tau_{xt} - \tau_x\tau_t + \tau\tau_{xxxx} - 4\tau_x\tau_{xxx} + 3\tau_{xx}^2 = 0$$

as a sufficient condition.

(c) Find the incoming and outgoing centre lines of each separated pulse and its net shift. Justify the two spatial derivatives in each moving-frame limit, rather than compare only pointwise values of τ.

The parameters $x_{j,0}$ describe the norming coefficients. They need not be actual minima positions during the interaction.

Classical Field Theory — Exercise Sheet 27

Matrix compatibility, formal charges and Bäcklund kinks

Questions only

The first two problems assume a smooth real KdV solution $u : I \times \mathbb{R} \to \mathbb{R}$, with I an open time interval, $u_t = 6uu_x - u_{xxx}$ and spatial period $L_{\mathrm{per}} > 0$. This is a conditional hypothesis, not an existence assertion on the circle. The last problem instead uses a real field on the whole line.

Exercise 27.1 (Converting the scalar KdV pair to matrices). For fixed $k \in \mathbb{C} \setminus \{0\}$ consider

$$-\psi_{xx} + u\psi = k^2\psi, \qquad \psi_t = -4\psi_{xxx} + 6u\psi_x + 3u_x\psi.$$

Set $\Psi = (\psi, \psi_x)^{\mathsf{T}}$.

(a) Eliminate higher spatial derivatives to derive $\Psi_x = U\Psi$, $\Psi_t = V\Psi$. Compute every entry of $U_t - V_x + [U, V]$ for an arbitrary smooth u.

(b) For the assumed periodic KdV field, state exactly what Lecture 27's monodromy theorem proves for these matrices. Does it establish the Poisson commutativity or independence of the resulting traces?

Exercise 27.2 (Local charges from a formal Riccati series). On a patch with $\psi \neq 0$, the scalar equation suggests $\rho = \psi_x/\psi$ and $\rho_x + \rho^2 = u - k^2$. In this problem work purely with formal Laurent series: put $\zeta = 2ik$ and seek

$$\rho = \frac{\zeta}{2} + \sum_{j \geq 1} a_j[u]\zeta^{-j}.$$

Equality means equality of each coefficient; convergence and a globally nonvanishing actual ψ are not assumed.

(a) Find the recursion and the first five coefficients. Integrate them over one spatial period and identify the first three nonzero integrals.

(b) Prove the formal identity $\rho_t = \partial_x[-u_x + (2u + 4k^2)\rho]$ under KdV. Conclude coefficientwise conservation. Explain which analytic conclusion about an actual spectral expansion has not been proved.

Exercise 27.3 (Bäcklund transformation and a sine-Gordon kink). On $\mathbb{R}^{1,1}$ use $\xi = (x+t)/2$, $\eta = (x-t)/2$. For two real C^2 fields $\phi, \widetilde{\phi}$, put $\alpha_B = (\phi + \widetilde{\phi})/2$ and $\beta_B = (\phi - \widetilde{\phi})/2$. For a fixed $a_B > 0$, the proposed *Bäcklund system* is

$$\partial_\xi \alpha_B = a_B \sin\beta_B, \qquad \partial_\eta \beta_B = a_B^{-1} \sin\alpha_B.$$

(a) Prove that any such pair solves $\phi_{\xi\eta} = \sin\phi$ and $\widetilde{\phi}_{\xi\eta} = \sin\widetilde{\phi}$.

(b) Put $\widetilde{\phi} = 0$ and solve for the branch $0 < \phi < 2\pi$ at finite points. Convert it to physical coordinates and find its velocity.

(c) For density $L = \frac{1}{2}(\phi_t^2 - \phi_x^2) - (1 - \cos\phi)$, compute the kink charge $Q = [\phi(+\infty) - \phi(-\infty)]/(2\pi)$, energy and lower-index momentum integral $\int \phi_t\phi_x \mathrm{d}x$. This line solution is not a periodic field for the monodromy theorem.

Classical Field Theory — Exercise Sheet 28

Tensor variation, gravitational waves and boundary action

Questions only

Use Lecture 28's curvature, inverse-metric and action conventions. All metrics, variations and coordinate maps are smooth. Repeated spacetime indices run from zero to three. Write $\eta = \mathrm{diag}(1,-1,-1,-1)$ only for the flat background. None of the problems asserts global existence for Einstein evolution.

Exercise 28.1 (The full Palatini tensor calculation). Let g_s be a smooth family of Lorentz metrics in fixed coordinates and put

$$k^{\mu\nu} = \partial_s g_s^{\mu\nu}\big|_0, \qquad h_{\mu\nu} = \partial_s (g_s)_{\mu\nu}\big|_0 = -g_{\mu\alpha}g_{\nu\beta}k^{\alpha\beta}, \qquad C^{\rho}_{\mu\nu} = \delta\Gamma^{\rho}_{\mu\nu}.$$

Here ∇ is the background Levi–Civita connection.

(a) Prove that C is a tensor of type $(1,2)$ and derive its expression in terms of ∇h. Do not assume that connection coefficients themselves are tensor components.

(b) Vary the full coordinate Riemann formula. Expand both covariant derivatives of C, including all its index terms, to prove $\delta R^{\rho}{}_{\sigma\mu\nu} = \nabla_\mu C^{\rho}_{\nu\sigma} - \nabla_\nu C^{\rho}_{\mu\sigma}$. Contract to obtain the Palatini identity without setting $\Gamma = 0$.

(c) Contract again to derive the inverse-metric boundary vector $v^\mu = \nabla^\mu(g_{\alpha\beta}k^{\alpha\beta}) - \nabla_\nu k^{\mu\nu}$. This is the complete tensor argument promised by Lecture 28.

Exercise 28.2 (Vacuum waves, gauge and the two polarisations). Set $g_{\mu\nu} = \eta_{\mu\nu} + s h_{\mu\nu} + O(s^2)$. Raise perturbation indices with η, write $h = \eta^{\mu\nu}h_{\mu\nu}$ and define $\bar{h}_{\mu\nu} = h_{\mu\nu} - \eta_{\mu\nu}h/2$.

(a) Derive $\Gamma^{(1)}$, $\mathrm{Ric}^{(1)}$ and $G^{(1)}$. Show explicitly that $G^{(1)}_{00}$ and $G^{(1)}_{0i}$ contain no second time derivatives of the perturbation. These are the linearized vacuum constraint equations on a slice.

(b) Under the covariant-metric Lie variation $h_{\mu\nu} \mapsto h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, derive the trace-reversed change and its divergence. In harmonic gauge $\partial^\mu\bar{h}_{\mu\nu} = 0$, derive the vacuum wave equation. No analytic theorem establishing that gauge globally is assumed or requested.

(c) For a mode $h_{\mu\nu} = \epsilon_{\mu\nu}e^{-i\nu t + i\nu x^3}$ with $\nu > 0$, impose harmonic gauge. Use residual null-wave gauge parameters to set the four temporal amplitudes to zero and derive the complete transverse traceless matrix. Count the independent polarisations. The real field is obtained by taking real parts; the plane mode itself is not claimed to have finite total energy on $\mathbb{R}^3$.

Exercise 28.3 (The non-null boundary action). Let Σ be a smooth non-null boundary patch of a region D. All variations have compact support in a collar of the interior of this patch and vanish near its edges and every other boundary piece. A Gaussian collar for the background may be used as a geometric given:

$$g = \epsilon_n \, \mathrm{d}n^2 + \gamma_{ab}(n, y)\mathrm{d}y^a\mathrm{d}y^b, \qquad \Sigma = \{n = 0\}, \quad D \text{ locally has } n \leq 0, \quad \epsilon_n = \pm 1.$$

The outward unit normal is $N = \partial_n$. Write $\mathrm{vol}_\Sigma = \iota_N \mathrm{vol}_g$ with outward-first orientation, $K = \nabla_\mu N^\mu$, and fix only the induced metric: $h_{ab}|_\Sigma = 0$ for $h_{\mu\nu} = \delta g_{\mu\nu}$. Normal components of h are initially unrestricted.

Gaussian normal coordinates for a smooth non-null hypersurface are the local construction described in Wald, *General Relativity*, §3.3, pp. 42–43. The boundary-action discussion is Appendix E.1, pp. 458–459. Use the explicit conventions in this question, including ϵ_n.

(a) Construct a vector field ξ vanishing on Σ such that $\widetilde{h} = h + \mathcal{L}_\xi g$ has $\widetilde{h}_{nn} = \widetilde{h}_{na} = 0$ near the patch. Derive and solve the required first-order normal equations. Explain why the induced variation is still zero.

(b) In this representative compute the normal flux of the Palatini vector and the variation of $2\epsilon_n \int_\Sigma K \mathrm{vol}_\Sigma$. Prove their cancellation.

(c) Use diffeomorphism invariance and Lecture 28's Bianchi identity to extend that result to the original arbitrary normal components. State the resulting gravitational action with the frozen overall sign. Specify why this is not a null-boundary or corner/joint formula.